



**SCHOOL OF COMPUTER SCIENCE
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CAT405 Intelligent Computing Major Project

System Requirements and Designs

PenangLens: Your Intelligent Travel Companion

LIM TING JUIN

163634

Supervisor: Assoc. Prof. Dr. Gan Keng Hoon

Examiner: Dr. Nur Intan Raihana Ruhaiyem

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DECLARATION

“I declare that the following is my own work and does not contain any *unacknowledged* work from any other sources. This report was undertaken to fulfill the requirements of the Undergraduate Major Project for the Bachelor of Science in Computer Science (Honors) program at Universiti Sains Malaysia”.

Signature : 

Name : Lim Ting Juin

Date : 2/12/2025

ABSTRAK

Pelancong moden yang meneroka destinasi yang kaya dengan budaya seperti Pulau Pinang sering berhadapan dengan cabaran lambakan maklumat dan pengalaman interaktif yang terhad semasa menerokai pelbagai tarikan di situ. Panduan tradisional dan alatan digital sedia ada kebiasaannya hanya menyediakan maklumat umum yang gagal memenuhi permintaan yang semakin meningkat untuk pengalaman penerokaan yang diperibadikan dan imersif. Jurang antara maklumat yang tersedia dan penglibatan yang bermakna ini boleh menjurus kepada pemahaman warisan budaya yang cetek, menyebabkan pelawat terlepas pandang konteks sejarah yang kaya serta kisah-kisah di sebalik sesuatu mercu tanda. Bagi menangani masalah ini, projek ini mencadangkan pembangunan sebuah aplikasi mudah alih inovatif berasaskan Kepintaran Buatan (AI) yang dinamakan "PenangLens". PenangLens bermatlamat untuk merevolusikan cara pelancong berinteraksi dengan pelbagai tarikan di Pulau Pinang, daripada tapak warisan UNESCO hingga ke landskap semula jadi dan seni jalanan yang meriah, dengan menggunakan teknologi penglihatan komputer termaju dan pemprosesan bahasa tabii. Objektif utama adalah untuk membangunkan sistem yang bukan sahaja mampu mengenal pasti mercu tanda daripada foto pengguna, malah berfungsi sebagai ejen AI proaktif yang boleh menjana itinerari perjalanan yang diperibadikan berdasarkan permintaan dalam bahasa tabii. Hasil yang dijangkakan ialah sebuah aplikasi mudah alih berfungsi yang bertindak sebagai teman perjalanan pintar, mentransformasikan pengalaman penemuan di lokasi serta perancangan pra-perjalanan bagi memperdalam pemahaman dan penghargaan pelancong terhadap segala keunikan yang ditawarkan oleh Pulau Pinang.

Kata Kunci: Pulau Pinang, Pelancongan, Kepintaran Buatan, Aplikasi Mudah Alih, Warisan Budaya, Penglihatan Komputer

ABSTRACT

Modern tourists exploring culturally rich destinations like Penang often face challenges of information overload and limited interactive experiences when discovering its wide array of attractions. Traditional guides and existing digital tools usually provide generic information that fails to meet the growing demand for personalized and immersive exploration. This gap between available information and meaningful engagement can lead to a superficial understanding of cultural heritage, causing visitors to miss the rich historical contexts and stories behind landmarks. To address this, the project proposes an innovative AI-powered mobile application called "PenangLens." PenangLens aims to revolutionize how tourists engage with Penang's diverse points of interest, from its UNESCO heritage sites to its natural landscapes and vibrant street art, by using advanced computer vision and natural language processing. The objective is to create a system that not only identifies landmarks from user photos but also functions as a proactive AI agent that can generate personalized travel itineraries based on natural language requests. The expected outcome is a functional mobile app that acts as an intelligent travel companion, transforming both on-the-spot discovery and pre-travel planning to deepen tourists' understanding and appreciation of all that Penang has to offer.

Keywords: Penang, Tourism, Artificial Intelligence, Mobile Application, Cultural Heritage, Computer Vision

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LIST OF ABBREVIATIONS AND SYMBOLS

AI	Artificial Intelligence
API	Application Programming Interface
BFF	Backend-for-Frontend
CAGR	Compound Annual Growth Rate
CMS	Content Management System
CNN	Convolutional Neural Network
COCO	Common Objects in Context (Dataset)
DINOv2	Self-supervised Vision Transformer (by Meta)
ERD	Entity Relationship Diagram
FR	Functional Requirement
GDP	Gross Domestic Product
GPS	Global Positioning System
GPU	Graphics Processing Unit
HITL	Human-in-the-Loop
IDE	Integrated Development Environment
JSON	JavaScript Object Notation
k-NN	k-Nearest Neighbors
LLM	Large Language Model
mAP	Mean Average Precision
MLLM	Multimodal Large Language Model
MVP	Minimum Viable Product
OAuth	Open Authorization
POC	Proof of Concept
POI	Point of Interest
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer

SDG	Sustainable Development Goal
SPPF	Spatial Pyramid Pooling - Fast
SSD	System Sequence Diagram
SWOT	Strengths, Weaknesses, Opportunities, and Threats
UGC	User-Generated Content
UI/UX	User Interface / User Experience
UML	Unified Modeling Language
UNESCO	United Nations Educational, Scientific and Cultural Organization
ViT	Vision Transformer
WBS	Work Breakdown Structure
YOLO	You Only Look Once (Object Detection Model)

1. INTRODUCTION

1.1. Project Background

Penang holds a distinguished position as a UNESCO World Heritage Site, with George Town receiving this prestigious designation on July 7, 2008, alongside Melaka, forming the "Melaka and George Town, Historic Cities of the Straits of Malacca" [1]. The 109.38-hectare core zone and 150.04-hectare buffer zone of the George Town UNESCO World Heritage Site contain a total of 5,013 buildings, representing an exceptional example of multicultural trading town in East and Southeast Asia [2]. This UNESCO recognition was granted under criteria (ii), (iii), and (iv), acknowledging George Town's outstanding universal value in demonstrating cultural exchange, bearing testimony to a cultural tradition, and representing an outstanding example of architectural ensemble [3]. This unique cultural and historical heritage is not just a point of pride; it is the primary engine of the state's tourism economy.

Fuelled by this heritage status, the tourism sector in Penang serves as a crucial economic pillar, contributing significantly to the state's Gross Domestic Product (GDP). According to the Penang Tourism Master Plan (2021-2030), the service sector contributes 49% of the state's GDP and has emerged as the primary driver of economic growth [4]. In recent years, Penang's tourism industry has demonstrated remarkable performance, with substantial visitor growth in 2024, including a staggering 218.86% increase in Chinese tourist arrivals and a 51.43% increase in Indian visitors [5, 6]. This growth trajectory reflects Penang's appeal as a globally recognized tourism destination. However, navigating the sheer density of these attractions presents a significant logistical challenge for tourists, who often struggle to create efficient and personalized travel plans on the fly. Sustaining this growth trajectory in a competitive global market requires not only preserving the heritage but also innovating the way it is presented to an increasingly digital-native audience.

Simultaneously, the global tourism industry is undergoing a profound digital transformation. The artificial intelligence (AI) market in tourism is projected to grow from USD 3.37 billion in 2024 to USD 13.87 billion by 2030, representing a

compound annual growth rate (CAGR) of 26.7% [7]. Over 80% of tourism executives reported increasing their AI budgets in 2024, indicating widespread industry recognition of AI's tremendous potential in enhancing travellers' experiences [8]. Despite this technological shift, a clear opportunity gap exists. Current tools that tourists rely on for cultural heritage exploration such as traditional guidebooks, blog articles, or generic map applications often suffer from fragmented information, lack of interactivity, and an absence of personalization. These tools typically provide static and standardized content that fails to meet modern tourists' growing demands for depth, authenticity, and personalized experiences.

The 'PenangLens' project aims to bridge this gap by developing an intelligent mobile application that transforms ordinary sightseeing into an immersive journey. By leveraging AI technology, the system enables tourists to instantly obtain rich background stories and in-depth analyses of historical buildings, street art, and cultural landmarks through simple photography actions. To realize this vision, the project focuses on four key technical objectives: establishing a secure user personalization system, engineering an AI core for visual interpretation, constructing an intelligent agent for itinerary planning, and designing a comprehensive content administration platform. These technical components work in tandem to address the specific challenges detailed in the following section.

1.2. Problem Statements

Despite Penang's thriving tourism industry and the increasing prevalence of digital tools, modern tourists still face several core challenges when experiencing its rich cultural heritage:

1) Lack of Personalization (The "One-Size-Fits-All" Problem)

Maps to: Objective 1 (Personalized Tourist Experience Module)

Current digital tools typically treat every tourist the same, resulting in inefficient "one-size-fits-all" experiences. For instance, platforms like TripAdvisor present the same static 'Top 10' lists to every user, forcing a history enthusiast to sift through irrelevant reviews for shopping malls or modern cafes, rather than proactively recommending heritage sites that match their specific profile. There is a lack of persistent identity systems that capture a user's preferences (e.g., history vs. nature) across sessions, preventing the delivery of a truly tailored tourism experience from the moment they log in.

2) The Gap Between Visual Discovery and Contextual Understanding

Maps to: Objective 2 (Immersive AI Visual Discovery Module)

A major challenge in modern tourism is the disconnect between seeing a landmark and understanding its deeper meaning or function. Interactions are often limited to superficial "check-in photography" because existing tools focus on identification rather than interpretation. For example, while tools like Google Lens can successfully identify a landmark's name (e.g., "Penang Hill" or "Kek Lok Si Pagoda"), they fail to provide the deep contextual details within the user's view such as the unique architectural elements of a building. This leaves tourists with a simple label but without the engaging narrative that makes the destination worth visiting.

3) Information Overload and Inefficient Planning

Maps to: Objective 3 (Intelligent Itinerary & Travel Companion Module)

Tourists face a "paradox of choice" due to the high density of attractions and generic online information. Visitors struggle to answer complex logistical queries like, "*I have three hours near Armenian Street, what can I do?*", without manually piecing together data from scattered sources. With 40% of global travellers already seeking AI

solutions to manage information overload [9], there is a critical need for an intelligent agent and dynamic map that can instantly generate feasible itineraries based on real-time constraints.

4) The Scalability Barrier of High-Quality Guided Experiences

Maps to: Objective 4 (Smart Content Curation & Management Module)

While human guides provide the gold standard for storytelling, they are a scarce and unscalable resource. It is impossible to provide expert-led, personalized interpretation for every visitor given that Penang hosts millions of tourists annually, including 5.2 million domestic and 3 million international arrivals in 2024 [10]. Hence, a digital solution is required to scale high-quality tourist guidance to the mass market.

However, maintaining 'expert-led' accuracy across thousands of landmarks presents a massive data management challenge, requiring a centralized system to curate, verify, and distribute cultural knowledge without compromising historical accuracy.

1.3. Motivation

The motivation for this project stems from several key aspects:

1) Responding to Experiential Tourism Trends

Modern tourists, especially millennials and Generation Z increasingly seek authentic “experiential tourism” rather than simple sightseeing. Studies show that 68% of Gen Z travellers prefer adventure-based and culturally immersive vacations [11], while 70% of millennials and Gen Z plan trips that emphasize the journey as much as the destination [12]. They crave deeper emotional and intellectual engagement, pursuing transformative experiences over passive observation. PenangLens addresses this shift by turning tourists into active explorers who uncover personalized stories and meanings from their own perspectives. Even during trip planning, travellers now expect seamless, AI-driven support to design journeys that are both efficient and spontaneously tailored.

2) Leveraging Cutting-edge AI Technology Potential

Recent advances in computer vision (such as DINOv2, YOLO11) and Large Language Models (LLM) provide unprecedented opportunities to revolutionize tourism experiences. This project aims to apply these cutting-edge technologies to the tourism and cultural field, exploring their tremendous potential in providing real-time, accurate, and contextualized information, offering an innovative practical case for industry applications. The convergence of these technologies enables the creation of intelligent systems that can understand both visual content and generate meaningful narratives, opening new possibilities for contextual interpretation.

3) Enhancing Penang's Tourism Competitiveness

Facing growing competition from emerging regional destinations, Penang must continuously innovate to sustain its unique appeal [4]. PenangLens, a high-tech tourism tool, enhances existing offerings while attracting younger, tech-savvy travellers seeking cultural depth. By blending tradition with modernity, it supports Penang's vision as a smart tourism destination. Such technological differentiation is vital as destinations vie for the attention of sophisticated travellers who expect seamless digital integration throughout their journeys.

1.4. System Objectives

- 1) To develop a secure Personalized Tourist Experience Module that provides cross-platform mobile application functionality with user registration, login, and personalized preference settings to create tailored tourism experiences.
- 2) To construct an Immersive AI Visual Discovery Module that accurately identifies landmarks from user photos, then analyzes them for specific visual details, and leverages a Large Language Model (LLM) to generate engaging, context-aware narratives.
- 3) To implement an Intelligent Itinerary & Travel Companion Module that functions as an intelligent agent, using an AI Travel Agent Chat and Personalized Itinerary Engine to generate customized travel plans, while also providing an Interactive Discovery Map for real-time exploration.
- 4) To create a comprehensive Smart Content Curation & Management Module that provides tools for full content management, allows for the review of AI performance via user feedback, and includes an innovative AI-assisted workflow to assist content creation from external sources.

1.5. Proposed solutions



Figure 1 : Logo of PenangLens

To address the challenges outlined, this project proposes the development of **PenangLens**, an intelligent mobile application designed to function not only as a personalized, interactive tour guide but also as a proactive travel planner. The solution is built upon a modern, modular architecture that leverages state-of-the-art AI to transform how tourists **discover, plan, and engage** with Penang's cultural heritage. The system is divided into four primary functional modules, supported by robust microservices-based backend architecture.

1.5.1 System Decomposition

The system's functionality is segmented into four high-level modules, ensuring a clear separation of concerns and scalability. This modular design allows for independent development, testing, and maintenance of each core component of the application.

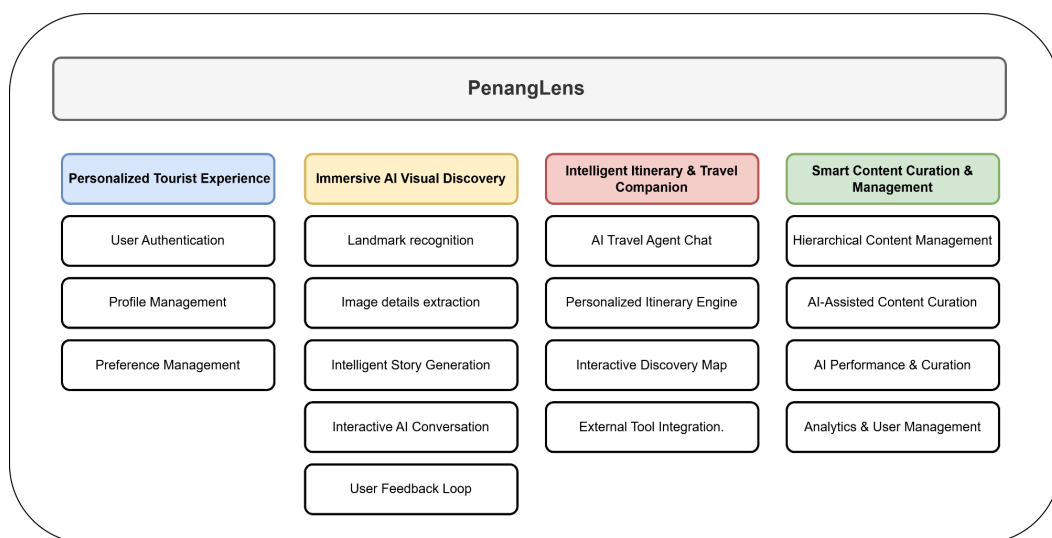


Figure 2: PenangLens' Overall module diagram

The system consists of the following core modules:

1. **Personalized Tourist Experience Module:** Manages secure user authentication (including Google login) and captures user interest tags (e.g., History, Nature, Food) during onboarding to create a persistent profile. This module drives the personalization engine, ensuring that recommendations are tailored to the user's specific tastes from the moment they log in.
2. **Immersive AI Visual Discovery Module:** The intelligent core that processes user photos to bridge the gap between seeing and understanding. It uses a multi-stage pipeline (DINOv2 and YOLO11) to identify diverse landmarks and specific architectural details, then leverages Retrieval-Augmented Generation (RAG) and Generative AI (LLM) to craft localized, context-aware stories.
3. **Intelligent Itinerary & Travel Companion Module:** Functions as a proactive travel agent and planner. It features a conversational AI chat interface and a dynamic Personalized Itinerary Engine that builds feasible, custom travel plans based on real-time constraints and preference, supported by an interactive map for on-the-go discovery.
4. **Smart Content Curation & Management Module:** A comprehensive web-based dashboard for administrators. It goes beyond basic data entry by utilizing AI-assisted tools with human verification to streamline content curation and includes a dedicated interface to monitor AI model performance through user feedback analytics. Uniquely, it manages the hierarchical structure of complex tourist sites, enabling the system to recognize that a single parent destination (e.g., Penang Hill contains multiple distinct child points of interest such as The Habitat, the Funicular Train).

1.5.2 System Architecture

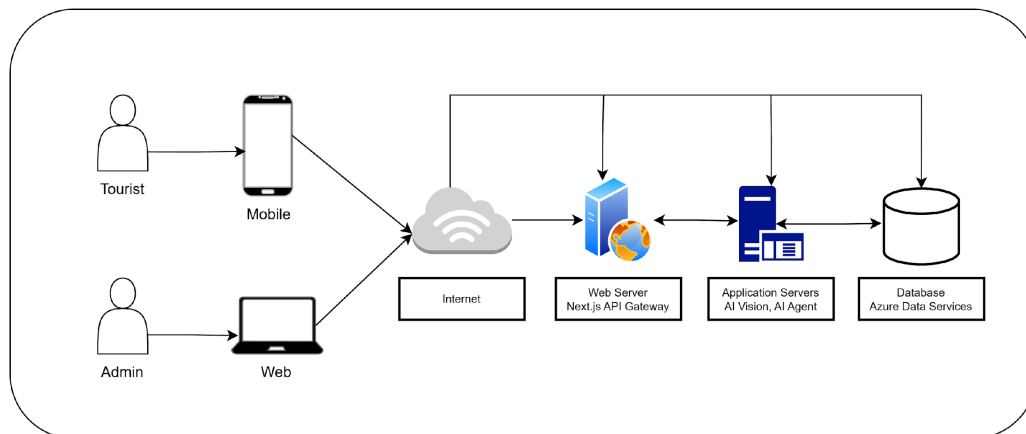


Figure 3: PenangLens' system architecture diagram

The system is designed using a modern, decoupled three-tier architecture to ensure scalability and a clear separation of concerns. This ensures scalability and allows the mobile client to remain lightweight while complex processing is offloaded to cloud.

The architecture supports two primary interfaces: a **Mobile Application** for tourists and a **Web Dashboard** for administrators. These clients interact with a central secure gateway which orchestrates requests. Computationally intensive tasks such as image recognition and itinerary generation are routed to specialized microservices deployed on cloud, while data is persisted in a scalable cloud database and high-performance vector store. This design ensures that high-demand AI processes do not impact the responsiveness of the user interface.

1.6. Benefits / Impact / Significance of Project

1) Benefits to Tourists

Significantly enriches and deepens tourism experiences, transforming static sightseeing into dynamic, interactive, and discovery-filled learning processes. It empowers tourists' curiosity for exploration and creates lasting, unique travel memories through personalized stories and seamlessly planned interactive itineraries.

2) Impact on Penang's Tourism Industry

Introduces an innovative high-tech tourism product to Penang, helping attract tourist groups seeking trendy and in-depth experiences. Additionally, by guiding tourists to focus on more diverse attractions and details, the application has the potential to distribute visitor flow from popular spots, promoting more sustainable tourism development patterns.

3) Significance for Promoting and Preserving Local Tourism & Heritage

PenangLens provides a completely new digital medium for destination storytelling and education. By transforming diverse local knowledge from historical facts to ecological insights into engaging interactive stories, it can popularize the value of Penang's unique offerings to broader audiences (particularly younger generations), thereby indirectly promoting social recognition for sustainable tourism preservation.

1.7. Uniqueness of solution

To establish the uniqueness of PenangLens, a comprehensive market analysis was conducted comparing it against existing tourism and cultural heritage applications. While several tools offer elements of travel guidance, none provide a deeply integrated, AI-driven narrative experience. Current solutions often function as static information repositories or basic identification tools, failing to create the dynamic, interactive dialogue that modern tourists increasingly seek [13].

Competitive Advantage

1) Deep Specialization and Personalization

Unlike broad platforms that cover global destinations, PenangLens is specialized, localized and laser-focused on the rich, dense collection of attractions in Penang. This focus enables a depth of content and a level of AI training that general-purpose applications cannot achieve. Moreover, its personalization engine actively filters content like maps and recommendations based on the user's declared interests, ensuring a more relevant and less overwhelming experience compared to the one-size-fits-all approach of most competitors.

2) From Reactive Tool to Proactive Companion

The most significant differentiator is the evolution from a static planner to an active travel companion. While conversational tools like Layla AI excel at generating text-based itineraries, their utility stops once the plan is made [14]. They cannot assist during the actual execution of the trip. PenangLens bridges this gap by integrating the itinerary with real-time visual discovery. The system not only builds a feasible plan in seconds but also "accompanies" the user throughout the journey, using its vision capabilities to identify and explain landmarks as the user encounters them. This creates a seamless loop where the digital plan adapts to and interprets the physical reality of the traveler, ensuring the user is never left with just a list of names but always has a guide to explain what they are seeing.

3) From Identification to Interpretation

While general-purpose visual tools like Google Lens can successfully perform basic landmark recognition [15], their analysis typically stops at providing a generic label or name (e.g., "Kek Lok Si Temple"). PenangLens' key innovation is its "coarse-to-fine" analytical ability. It first identifies the overall landmark and then goes a crucial step further by recognizing specific architectural details, symbols, or artistic elements within the context of the user's unique photograph. This dual-layer understanding allows it to move beyond simple identification into true visual interpretation.

4) From Static Content to Dynamic Dialogue

Traditional digital tools like TripAdvisor's "Trips" feature [16] or Audio Guide Apps offer pre-written text and pre-recorded audio. The information is static and unidirectional. PenangLens revolutionizes this paradigm by using its detailed visual analysis to dynamically generate unique stories for each user. Furthermore, the inclusion of an interactive AI chat transforms the user from a passive information consumer into an active participant in a dialogue, allowing them to probe deeper into topics of interesting experience that are currently unavailable in the market.

1.8. Organization of the Report

This report is structured to provide a comprehensive analysis and design of the proposed PenangLens system. The organization of the report is as follows:

Chapter 1: Introduction establishes the project context within Penang's diverse tourism landscape, addressing challenges like information overload and lack of personalization. It defines the objectives, motivation, and the proposed AI-powered mobile app solution, concluding with the project's scope, uniqueness, and contributions.

Chapter 2: Background and Related Work provides a theoretical framework for the technologies used, including computer vision and generative AI. It also presents a critical review of existing tourism applications and AI travel tools, highlighting their limitations and establishing the market gap that PenangLens aims to fill.

Chapter 3: System Requirements and Analysis details the project's scope, capabilities, and limitations. It outlines the development methodology and project management plan, followed by a detailed requirements analysis using Unified Modeling Language (UML) diagrams. The chapter concludes with a comprehensive design of the system architecture, the specific technologies deployed, and the project's alignment with Sustainable Development Goals (SDGs).

Chapter 4: Conclusion summarizes the project's contributions and discusses the expected outcomes of the development phase.

2. BACKGROUND AND RELATED PROJECT

2.1. Status of Project Development

PenangLens is a new greenfield initiative currently in the requirements analysis and system design phase. The system is being developed from the ground up to integrate advanced AI technologies, including DINOv2 and YOLO11 for computer vision and Gemini for Large Language Model (LLM) capabilities.

Unlike existing commercial applications that mainly provide static content or basic location-based services, PenangLens introduces a multi-stage AI analysis pipeline that combines visual recognition, detailed attribute analysis, and interactive storytelling. This approach bridges the gap between surface-level sightseeing and deeper contextual understanding, addressing a clear market need for an immersive and intelligent travel companion tailored specifically to Penang.

To clearly visualize the project's development goals, Table 1 outlines the shift from traditional methods to the proposed AI-driven capabilities:

Aspect	Current Market Approaches	PenangLens Innovation
Discovery Method	Manual text search or scanning QR codes physically located at sites.	Visual-First Discovery: Instant identification of landmarks and details using AI Computer Vision (DINOv2) followed by the granular detection of specific details like architecture (via YOLO11).
Information Delivery	Static, pre-written descriptions that are the same for every user.	Generative Storytelling: Dynamic, context-aware narratives generated by LLM (Gemini) based on input.
Planning Experience	Fragmented tools; users must manually piece together maps and blogs.	Autonomous Agent: A proactive AI agent that generates feasible itineraries based on natural language requests.

Table 1: Comparison of Traditional Travel Applications and PenangLens

2.2. Background Study

2.2.1. Smart Tourism and Experiential Travel

The concept of "Smart Tourism" refers to the integration of digital technologies into physical tourism infrastructure to enhance visitor experiences and operational efficiency. Unlike traditional tourism, which often relies on static, one-way communication (e.g., guidebooks or signage), smart tourism facilitates a dynamic ecosystem where travelers actively "co-create" their experiences through real-time interaction and personalized data [17].

Within this framework, there is a shift towards **Experiential Travel**, where modern tourists seek to engage deeply with a destination's narrative rather than merely visiting checkpoints. This suggests that digital tools enhance value co-creation by enabling more personalized, immersive, and efficient tourism experiences, allowing for dynamic interaction rather than passive consumption [18]. This capability allows destinations to increase their competitiveness by moving beyond traditional mass tourism models to offer the highly personalized contexts that modern travelers demand [19].

2.2.2. Computer Vision in Tourism: From Detection to Understanding

To effectively analyze the diverse visual landscape of a tourism destination, the system implements a **"Global-to-Local" hierarchical analysis pipeline**. This architectural pattern mimics human visual cognition where at first, the system establishes the **Identity** of the scene using global visual descriptors, and subsequently, it performs **Detection** to isolate fine-grained attributes using a specialized object detector. For example, after identifying the Kek Lok Si Pagoda, the system activates the detection pipeline trained specifically to distinguish fine-grained features of its architecture details.

1) Identification: Visual Information Retrieval (DINOv2)

The first phase identifies the landmark (e.g., *Kek Lok Si Pagoda*) through instance-level image retrieval using **DINOv2**. Unlike supervised models, DINOv2 is a foundation model trained via a **Student-Teacher distillation framework**, which

optimizes for feature consistency rather than label memorization [20]. This approach produces high-dimensional vector embeddings that are invariant to extreme lighting, occlusions, or camera angles [21]. Such robustness allows the system to perform a **Similarity Search** (k-NN) to retrieve the exact matching location from the database with high precision, even if the user's photo differs from the canonical reference image.

2) Detection: Context-Aware Grounding (YOLO11)

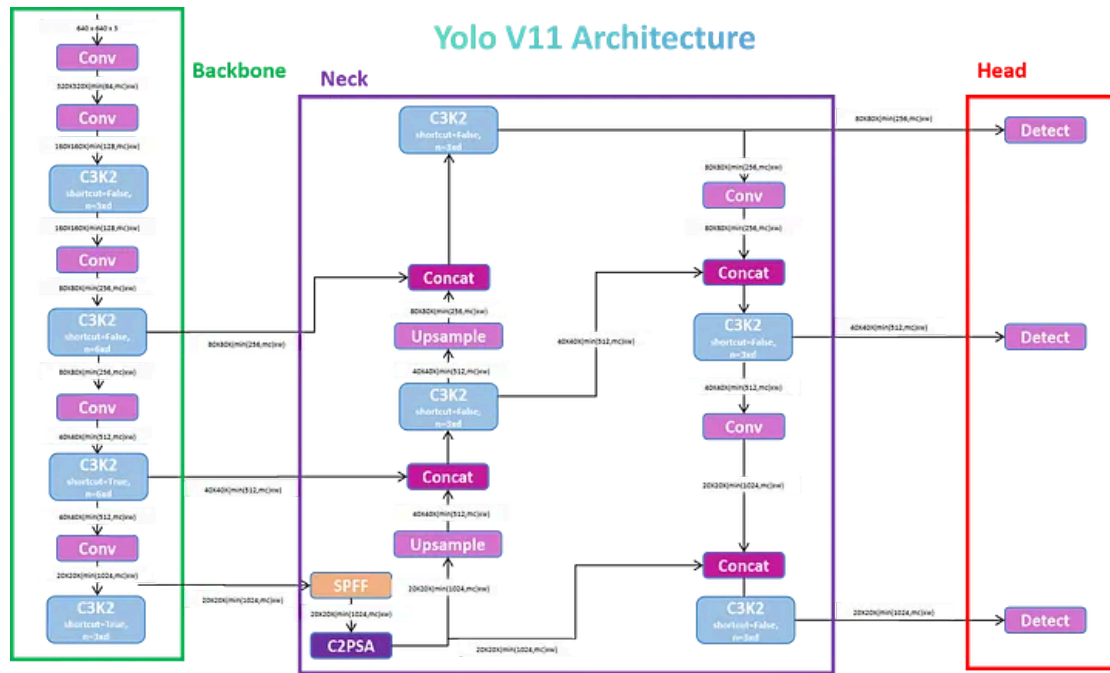


Figure 4: YOLO11 Model Architecture

Once the landmark is identified (e.g., Kek Lok Si), the system performs **granular analysis** to isolate specific architectural components (e.g., "Burmese spiral dome") using YOLO11, the latest iteration of the You Only Look Once family. Unlike the previous generation, YOLO11 introduces the **C3k2 block (Cross-Stage Partial Network)** and **SPPF (Spatial Pyramid Pooling - Fast)** mechanisms. These architectural upgrades allow the model to focus on intricate spatial features and overlapping geometries with significantly higher efficiency, achieving superior accuracy with approximately 22% fewer parameters than its predecessors [22].

To achieve the deep understanding required for heritage architecture while mitigating the risk of overfitting on a custom dataset, a **partial fine-tuning strategy** will be

employed. Although standard pre-trained models capture generic features, the "domain gap" between these generic objects and specific architectural tiers is substantial [23]. Therefore, this strategy utilizes the pre-trained weights of the YOLO11 Backbone to extract robust low-level visual features (edges and textures), while retraining the Neck and Head layers. This allows the model to adapt its high-level semantic understanding to the specific geometric nuances of Penang's architecture without destroying the foundational feature extractors learned from large-scale datasets.

2.2.3. Generative AI for Context-Aware Storytelling (Gemini LLM)

Generative AI, particularly Large Language Models (LLMs), has transformed tourism applications by enabling systems to act as virtual narrators capable of personalized, multilingual storytelling. However, standard LLMs can suffer from "hallucination," generating plausible but factually incorrect information [24].

To ensure accuracy in explaining the history of a landmark like Fort Cornwallis, **Retrieval-Augmented Generation (RAG)** is employed. RAG combines the generative fluency of an LLM (such as Google Gemini) with the factual precision of an external knowledge base. When a query is received, the system first retrieves verified facts (e.g., dates, names, history) from the vector database and feeds them into the LLM as context [25].

2.2.4. Autonomous AI Agents and Cognitive Architectures

1) AI Agents

An Artificial Intelligence (AI) Agent is defined as a system that autonomously performs tasks by designing workflows with available tools [26]. Unlike traditional software or passive LLM chatbots that simply respond to user queries based on pre-trained patterns, AI Agents operate as active problem solvers. They utilize "tools" such as APIs, search engines, or function calls to interact with the external world.

A core paradigm driving modern agents is the ReAct (Reasoning + Acting) framework. In this framework, the model generates a "thought trace" to plan its next move, executes an action (e.g., checking a weather API), and then observes the output before deciding the next step [27]. This "cognitive loop" allows agents to handle complex, ambiguous tasks that require dynamic adjustment, rather than following a rigid script.

2) Autonomous AI Agents in Travel Planning using LangGraph

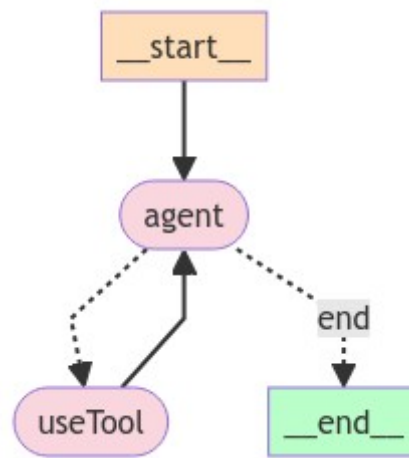


Figure 5: A cyclic agent workflow in LangGraph enabling self-correction

Travel planning is a cognitive task that requires gathering information from disparate sources (maps, distances, reviews & ratings) and synthesizing it into a logical schedule. Traditional tools force users to perform this aggregation manually. Autonomous Agents represent a paradigm shift, moving the system from a passive search tool to a proactive companion that manages logistical complexity.

To build robust agents for tourism, the project utilizes LangGraph, a framework that models agent workflows as directed cyclic graphs as shown in Figure 5. Unlike linear chains that follow a strict input-to-output sequence, LangGraph allows for loops and conditional logic [28]. This architecture enables self-correction, where the agent can "loop back" to critique and optimize its own plan. For instance, removing lower-priority stops if an itinerary exceeds a time limit, thereby mimicking the iterative reasoning of a human travel agent. Furthermore, the framework maintains a persistent state (memory) throughout the conversation, ensuring that specific user

constraints or interests like "prefer historical site" or "wheelchair access" are retained across the entire planning session [28].

2.3. Related Works and Comparison

To position PenangLens within the current tourism technology landscape, it is essential to analyze existing commercial applications. These platforms can be broadly categorized into **General Tourism Applications** (visual lookup and review aggregators) and emerging **AI-Powered Travel Tools** (generative itinerary planners).

2.3.1 Existing General Tourism Applications

2.3.1.1 Google Lens

Google Lens represents the industry standard for visual search, utilizing deep learning to identify objects, landmarks, and text through a mobile camera. It functions primarily as a visual lookup tool, linking physical objects to indexed web search results [15].

Strengths: Its primary strength is the vast scale of its training database, allowing it to recognize millions of global landmarks with high accuracy and speed. It effectively answers the question, "*What is this?*"

Limitations: However, Google Lens creates a "Semantic Gap" in tourism. It identifies the *label* of an object (e.g., "Kek Lok Si Pagoda") but fails to provide the *narrative context* or specific architectural breakdown (e.g., explain the pagoda's unique three layers design). It creates a disjointed experience where the user must leave the camera interface, perform web search themselves to read static web pages, breaking immersion. Furthermore, it lacks a personalization engine; the results shown are identical for every user regardless of their interest in history or art.

2.3.1.2 TripAdvisor

TripAdvisor operates as a massive aggregator of User-Generated Content (UGC), offering millions of reviews, photos, and pre-set tour listings. It serves as a reputation engine and planning repository [16].

Strengths: It excels at "Social Proof," helping users assess the quality of a destination based on peer reviews. Its database of opening hours and ticket prices is generally robust.

Limitations: The platform suffers from the "Paradox of Choice." The sheer volume of unstructured information forces users to manually sift through hundreds of reviews to piece together an itinerary. It is purely text-based and reactive; it cannot "see" what the user is looking at in real-time, nor can it dynamically adjust an itinerary based on the user's current location and remaining time.

2.3.1.3 Layla AI

The recent surge in Generative AI has led to tools like Layla AI, which utilize Large Language Models (LLMs) to generate textual travel itineraries based on simple prompts [14].

Strengths: This tool solves the "Cold Start Problem" in planning. A user can request a "3-day itinerary for Penang," and the system generates a schedule in seconds, significantly faster than manual research.

Limitations: However, these planners lack "Visual Grounding." They operate solely on text and do not integrate with the user's camera, functioning only as pre-trip planning utilities rather than on-trip discovery companions. Furthermore, without a robust retrieval-augmented (RAG) framework connected to a verified local database, these generic LLM wrappers are prone to "hallucinations," often recommending permanently closed restaurants or non-existent routes.

2.3.2 Related Academic Research

Beyond commercial applications, academic research has extensively explored the application of deep learning for cultural heritage preservation and smart tourism.

1) Computer Vision for Heritage Classification

Early research on cultural heritage analysis predominantly relied on supervised Convolutional Neural Networks (CNNs) for tasks such as surface defect detection and structural condition assessment. Fu and Angkawisittpan (2023) investigated deep learning methods for detecting surface defects on heritage buildings and demonstrated that transformer-based architectures, specifically a Swin Transformer integrated with YOLOv5, outperformed multiple CNN-based segmentation models across various defect categories [29]. Their findings indicate that transformer-based models offer more robust feature representations than CNN-based approaches, whose dependence on large labeled datasets limits scalability for under-documented heritage sites.

Besides, more recent research has shifted toward self-supervised vision foundation models to overcome these limitations. Oquab et al. introduced DINOv2, a self-supervised Vision Transformer that learns general-purpose visual representations without human annotation and achieves strong performance across a wide range of downstream vision tasks [30]. These results highlight the suitability of self-supervised ViT representations for fine-grained visual recognition problems common in cultural heritage analysis, particularly in data-scarce scenarios.

PenangLens adopts this modern approach, using the DINOv2 Vision Transformer for the initial Identification phase (Visual Information Retrieval) and then employing YOLO11 which integrates transformer-style attention mechanisms (C2PSA)—to enable robust, specialized detection of the heritage architectural assets.

2) Large Language Models (LLMs) for Itinerary Planning

In the domain of intelligent travel planning, Large Language Models (LLMs) have increasingly been explored as autonomous or semi-autonomous agents. Wang et al. provided a comprehensive survey of LLM-based autonomous agents, outlining common system architectures, planning mechanisms, memory designs, and evaluation

strategies [31]. The survey further identifies key challenges in deploying LLM agents in real-world applications, including planning reliability, consistency across multi-step tasks, and effective interaction with external tools and knowledge sources.

To mitigate these limitations, recent research has emphasized retrieval-augmented generation (RAG) and structured knowledge integration. Song et al. proposed the TravelRAG framework, which combines a multi-layer knowledge graph with an LLM to support tourist attraction retrieval and itinerary generation [32]. By grounding language generation in structured geographic and semantic data, TravelRAG improves relevance and reduces factual inconsistencies associated with standalone LLMs.

Building on this line of work, PenangLens implements a ReAct (Reason + Act) agent workflow using LangGraph, enabling the system to iteratively reason, query a local heritage database, and verify outputs. This approach reduces reliance on the LLM's parametric knowledge and enhances factual reliability in tourism recommendations.

3) Multimodal Large Language Models (MLLMs) for Tourism Assistance

Recent university-led research has demonstrated the effectiveness of multimodal systems that integrate vision and language models for cultural tourism assistance. Huang and Song introduced MACAVA, a multimodal AI assistant designed for cultural heritage tourism in Macau [33]. The system employs a hybrid visual recognition pipeline, combining image-text embedding models for initial visual retrieval with large language models for semantic interpretation and explanation. Their work illustrates the advantages of multimodal pipelines over unimodal vision-only approaches for cultural interpretation.

In addition, the MACAVA project explored the use of open-vocabulary visual detection models namely YOLO-World to identify specific architectural or symbolic elements within complex heritage scenes [33]. Although YOLO-World was originally considered, PenangLens ultimately pivots to the specialized, fully supervised YOLO11 model. This decision prioritizes maximum detection precision and speed for the fixed set of fine-grained architectural features over the flexible but less precise open-vocabulary capabilities, ensuring high-fidelity visual grounding for the subsequent LLM interpretations.

2.4. Features Comparison of Existing Systems

2.4.1. Comparison Matrix

To visualize the functional gap in the market, Table 2 compares PenangLens against key competitors across seven critical dimensions of the modern tourism experience.

Feature	PenangLens	Layla AI	Google Lens	TripAdvisor	Audorista
AI Landmark Recognition	✓	✗	✓	✗	✗
Contextual Detail Recognition	✓	✗	✗	✗	✗
Dynamic Story Generation	✓	✗	✗	✗	✗
Interactive AI Chat	✓	✓	✗	✗	✗
Specialized Heritage Focus	✓	✗	✗	✓	✓
Personalization Engine	✓	✓	✗	✗	✗
AI Itinerary Planning	✓	✓	✗	✗	✗

Table 2: Comparison of PenangLens with other competitors

2.4.2. Critical Analysis

The comparison matrix highlights three distinct clusters of limitations in the current market:

1. **The "Vision-Only" Cluster (Google Lens):** These tools excel at recognizing objects but lack the generative capability to tell stories or the planning capability to suggest what to do next. They function as isolated identification utilities rather than guides.
2. **The "Text-Only" Cluster (TripAdvisor, Layla AI):** These platforms are strong on data and planning but are disconnected from the physical visual reality of the tourist. They cannot interpret a specific mural, architectural detail, or natural feature standing in front of the user.
3. **The "Static Content" Cluster (Audorista/Audio Guides):** While these offer specialized local focus, the content is pre-recorded and linear. They cannot answer follow-up questions or adapt the narrative to the user's specific interests.

PenangLens addresses these gaps by converging these capabilities. It is the only system proposed to integrate **Visual Recognition** (DINOv2/YOLO11) with **Generative Storytelling** (Gemini) and **Agentic Planning** (LangGraph) in a single application. Unlike competitors that force users to switch between a 'camera app' and a 'planning app,' PenangLens allows users to seamlessly transition from identifying a landmark to asking the AI Agent for travel advice, effectively consolidating the entire tourist journey into one platform.

2.5. Introduction of Proposed Work

PenangLens integrates DINOv2, YOLO11, Google Gemini, and LangGraph to create a unified intelligent tourism platform. By synthesizing the findings from the literature review and market analysis, the project identifies key methodologies to adopt and critical gaps to address.

The **strengths** of existing tourism technologies, which are validated by the market and will be **maintained** in the proposed system, include:

1. **Visual Search Interaction:** The intuitive "point-and-shoot" discovery model is retained as the primary entry point for on-the-go exploration.
2. **Conversational Interfaces:** Natural language query capabilities are maintained to simplify information retrieval, removing the barrier of complex search filters.
3. **Map-Based Exploration:** Geospatial visualization of attractions provides essential navigation context and will be preserved to guide users through the physical space.

On the other hand, the following **areas for improvement** are identified from the limitations of current systems, which PenangLens addresses:

1. **From Labeling to Interpretation:** Moving beyond generic labels (e.g., "temple"), the system utilizes **YOLO11** and **RAG** to identify specific architectural details and provide deep, context-aware narratives.
2. **From Hallucination to Accuracy:** To mitigate the risk of fictitious information in standard LLMs, the system enforces factual accuracy by grounding generative outputs in a verified local vector database.

3. **From Static to Agentic Planning:** Addressing rigid itineraries, the system employs LangGraph Agents to reason and self-correct plans in real-time (e.g., dynamic rerouting), mimicking the adaptability of a human guide.
4. **Unified Workflow:** The solution consolidates visual discovery, storytelling, and itinerary planning into a single application, eliminating the need for users to fragment their attention across multiple apps.

This project aims to bridge these gaps by shifting the paradigm from disparate tools to an integrated intelligent companion.

3. SYSTEM REQUIREMENTS / ANALYSIS

3.1. Detailed Introduction of Proposed Work

3.1.1. Project Scope & Modules

3.1.1.1 Personalized Tourist Experience Module

This module manages secure user onboarding and user personalization recommendation engine. This includes handling the identity management, authentication and token security to ensure safe communication between the mobile client and the API gateway.

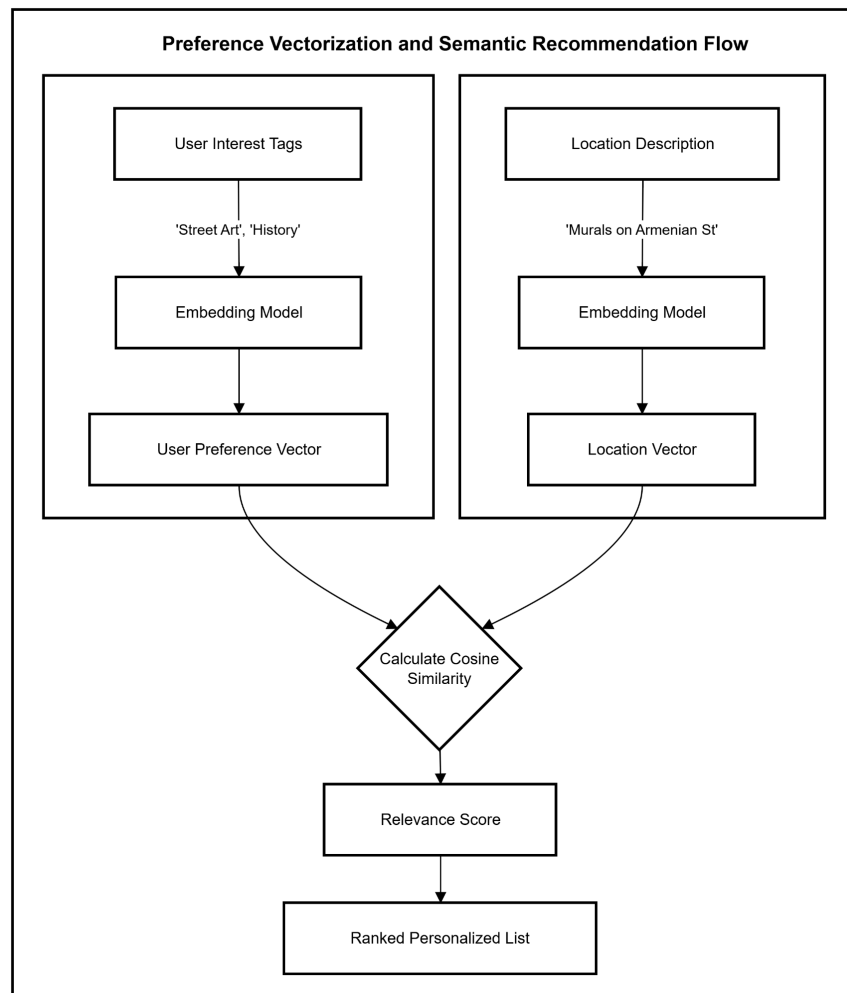


Figure 6: Preference Vectorization and Semantic Recommendation Workflow

As illustrated in Figure 6, the recommendation engine operates on a semantic vector space rather than simple keyword matching.

1. **User Profile Vectorization:** User interest tags (e.g., 'Street Art', 'History') are aggregated and processed by an embedding model to generate a normalized User Preference Vector.
2. **Location Vectorization:** Simultaneously, the descriptive text of each location is embedded into a Location Vector within the same high-dimensional space.
3. **Semantic Matching:** The system calculates the Cosine Similarity between the user vector and each location vector. This mathematical operation determines the semantic closeness of a user's interests to a location's characteristics.
4. **Ranking:** Locations are assigned a relevance score based on this similarity, and the final output is a personalized list ranked from highest relevance to lowest, ensuring recommendations align with the user's intent even without exact keyword overlaps.

3.1.1.2 Immersive AI Visual Discovery Module (The Vision Pipeline)

This module functions as the primary intelligent input, transforming raw visual data into structured narrative context. The recognition process follows a "Coarse-to-Fine" Hierarchical Pipeline consisting of three distinct stages: **Global Retrieval (Identify)**, **Supervised Detail Detection (Detect)**, and **Generative Synthesis (Synthesize)**.

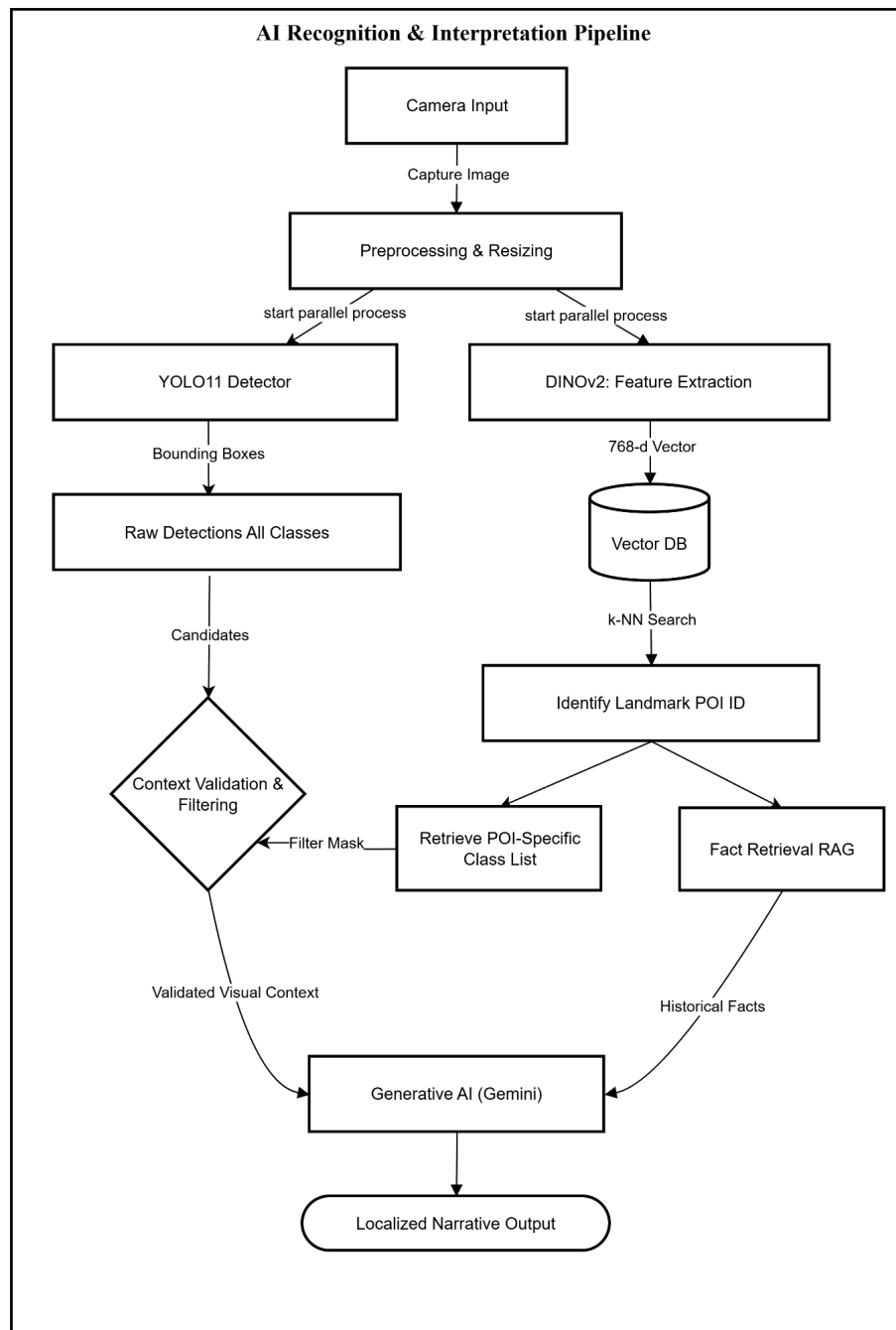


Figure 7: The "Coarse-to-Fine" AI Vision Pipeline Workflow

a. Global Visual Retrieval (DINOv2)

The first phase identifies the landmark identity (e.g., "Penang Hill Funicular Train") from the user's photo. The system utilizes the **DINOv2 ViT-B/14 (Base)** model as the primary feature extractor. This backbone is selected for its optimal balance between inference latency and retrieval accuracy, enabling scalable cloud deployment.

- **Vector Extraction:** The input image is processed to extract a 768-dimensional feature vector. Unlike supervised models that rely on class labels, this self-supervised backbone captures robust geometric and semantic features invariant to lighting conditions and occlusions.
- **Retrieval Mechanism:** The extracted vector is queried against a **Vector Database (Azure AI Search)** containing pre-computed embeddings of verified Penang landmarks. A K-Nearest Neighbors (k-NN) search algorithm with cosine similarity is employed to identify the closest matching landmark ID.

b. Fine-Grained Detail Detection (YOLO11)

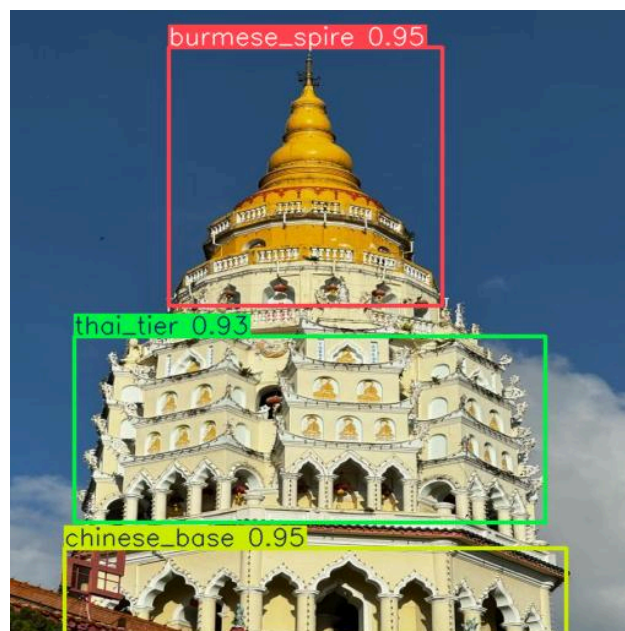


Figure 8 : Example of Fine-Grained Detection Output of Kek Lok Si Pagoda

The detection phase uses the **YOLO11** architecture for high-precision, fine-grained object detection. This stage, executed **in parallel** with the DINOv2 retrieval, employs a fully supervised model to detect specific heritage features.

- **Transfer Learning Strategy:** The model is specialized through **Transfer Learning** using our custom-curated dataset. This ensures the model accurately recognizes the precise architectural details (e.g., "Chinese Base," "Burmese Spire" as shown in Figure 8) which is relevant to the landmark's context.
- **Inference Workflow (Parallel Execution with Sequential Filtering):** To minimize latency, the DINOv2 retrieval process and the YOLO11 detection inference are executed **in parallel** on the input image. The system then implements a sequential filtering step to ensure contextual accuracy:
 1. **Parallel Execution:** Both the **DINOv2** retrieval (identifying the POI ID) and the **YOLO11** model's inference (generating raw detections) are executed concurrently.
 2. **Retrieval of Class List:** The **Landmark POI ID** (from DINOv2) is used to retrieve a **POI-Specific Class List** from the database.
 3. **Sequential Validation:** A **Validation** component then combines the POI-Specific Class List and the YOLO11 raw detections. It performs a logical intersection to retain only the objects that are valid for that specific POI. This final filtering step eliminates false positives, maximizes detection confidence, and yields the validated visual context.

c. Generative Synthesis (Synthesize)

The final phase uses the validated visual context and retrieves historical knowledge to synthesize a localized, engaging narrative for the user. This process leverages a **Retrieval-Augmented Generation (RAG)** pipeline anchored by a foundation Large Language Model (LLM), **Google Gemini**.

- **Fact Retrieval (RAG):** The validated visual context (Landmark POI ID + Detected Details) acts as the query. The system performs a targeted retrieval against a proprietary knowledge base to fetch verified, factually grounded facts.
- **Context Prompt Construction:** A structured prompt is assembled for the LLM. This prompt explicitly contains:
 - 1. The confirmed Landmark Name
 2. The list of visually **Detected Details** (e.g., "Chinese Octagonal Base"),
 3. The **Retrieved Facts** from the Knowledge base.
- **Narrative Generation:** The LLM receives the comprehensive context prompt and is instructed to generate a concise, engaging narrative tailored to the user's detected features and location.

3.1.1.3 Intelligent Itinerary & Travel Companion Module (The Agent)

This module operates as the central intelligent hub for travel logistics, moving beyond static itinerary templates by utilizing **Autonomous AI Agents**. Unlike traditional chatbots that simply respond to text, this system is engineered to actively reason, plan, and self-correct using a graph-based cognitive architecture.

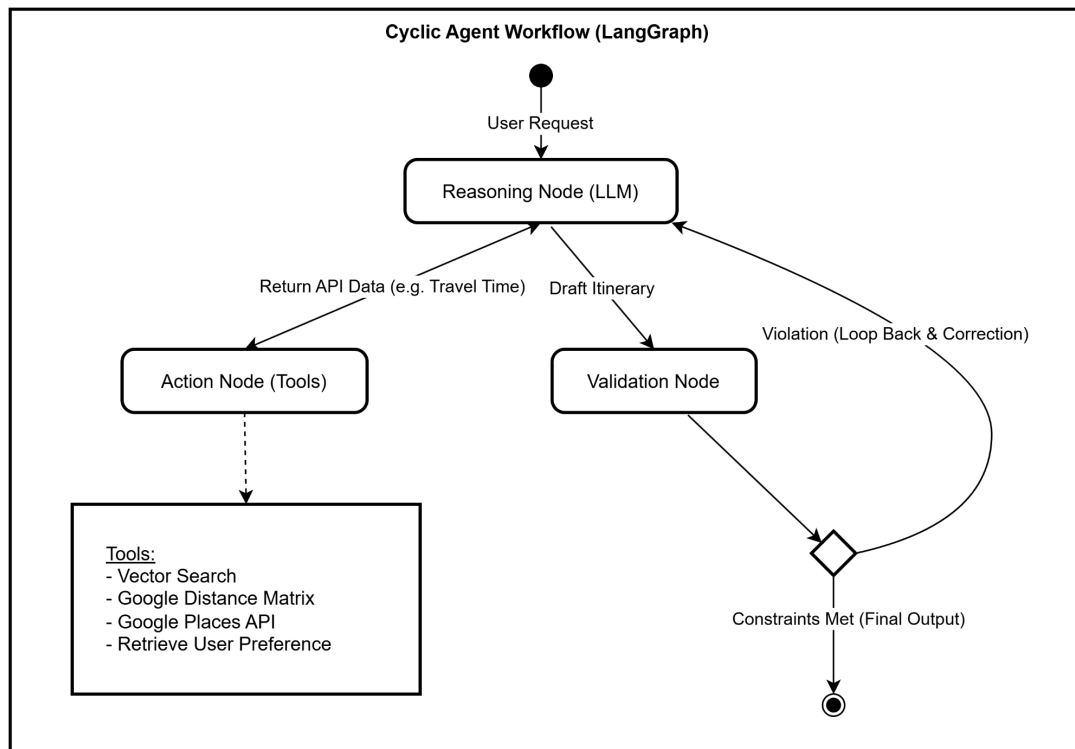


Figure 9: Cyclic Agent Workflow (LangGraph)

1) Cognitive Architecture: Cyclic Graph Workflow

The planning engine is built upon the LangGraph framework, which models the itinerary generation process as a Directed Cyclic Graph. This architecture is superior to linear chains (Input -> Output) because it enables a "Cognitive Loop" where the agent can critique its own work and refine the plan iteratively before presenting it to the user.

2) The ReAct Pattern (Reason + Act)

The agent implements the ReAct (Reasoning + Acting) paradigm to solve complex logistical problems. When a user submits a request (e.g., "Plan a 4-hour heritage walk starting at 9 AM"), the system does not hallucinate a schedule. Instead, it executes a rigorous logic flow:

- I. **Reasoning Phase:** The agent parses the request to identify constraints:
Duration (4h), Theme (Heritage), Start Time (09:00).
- II. **Action Phase (Tool Use):** The agent triggers external tools via function calling:
 - A. **Retrieve User Preference:** Fetches the user's personalization vector (Module 1) to ensure recommendations align with specific interests like "History" or "Nature".
 - B. **Vector Search:** Queries the system database to retrieve specific landmarks (e.g., "Heritage sites") that match the semantic intent of the user's request.
 - C. **Google Places API:** Validates real-time operational data, such as opening hours and operating status, to prevent recommending closed venues.
 - D. **Google Distance Matrix:** Calculates realistic travel times (walking or driving) between the identified candidates to optimize the route sequence.
- III. **Observation Phase:** The agent receives the data (e.g., "Walking from Fort Cornwallis to Pinang Peranakan Mansion takes 12 mins").
- IV. **Synthesis Phase:** The agent constructs the final timeline, ensuring travel buffers are included.

3) Constraint Validation & Self-Correction

A critical innovation in this module is the **Validation Node**. Before finalizing an itinerary, the agent checks the draft against hard constraints (e.g., "Is the museum open today?"). If a violation is detected, the graph "loops back" to the Reasoning Node to swap the closed venue for an alternative, ensuring the final output is always logically feasible.

3.1.1.4 System & Content Administration Module

This web-based dashboard serves as the command center for the PenangLens ecosystem, providing administrators with tools to manage the hierarchical structure of heritage data and monitor system health. Unlike traditional CMS platforms that rely heavily on manual data entry, this module integrates an AI-Assisted Curation Pipeline to solve the "Cold Start" problem of populating a large-scale tourism database.

1) Hierarchical Content Management

The system implements a relational schema that supports parent-child relationships for complex landmarks. For example, a parent entity "Kek Lok Si Temple" can house multiple child POIs (Points of Interest) such as "Pagoda of Rama VI" and "Guan Yin Statue." This granular structure allows the AI vision model to identify specific components of a site rather than just the general location.

2) Ingestion & Indexing Pipeline (Technical Strategy)

To ensure the highest standard of data integrity for heritage facts, the system employs a **Human-in-the-Loop (HITL)** architecture. This approach combines expert verification with automated backend processing to generate AI-ready assets instantly upon submission.

- **Automated Visual Indexing:** Upon the upload of a reference image, the system triggers an asynchronous background job using the **DINOv2** model. This process extracts and indexes the image's feature vector in the **Azure AI Search** database, rendering the landmark immediately recognizable to the mobile app without requiring model retraining.
- **Semantic Text Embedding:** As administrators input historical descriptions, the system automatically generates text embeddings. This powers the "Semantic Search" capabilities within the user's itinerary planner.

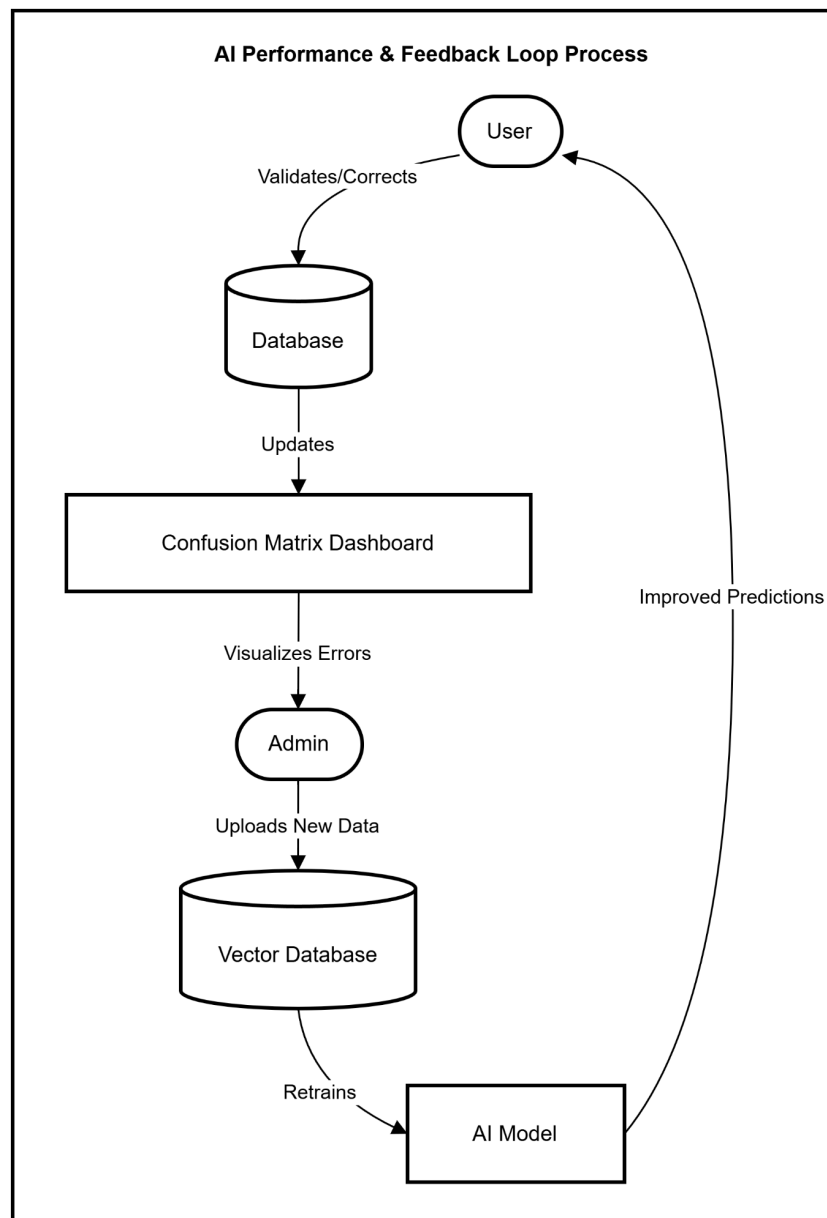


Figure 10: AI Performance & Feedback Loop Process

3) AI Performance & Feedback Loop

The administration module includes a specialized **Confusion Matrix Dashboard** to monitor the computer vision model's accuracy. It visualizes data from the "User Feedback Loop" submitted from Module 2, highlighting specific classes where the model frequently made mistakes (e.g., confusing "Clan Jetties" with "Chew Jetty"). This allows admins to identify "weak" landmarks and upload better reference images.

- **Immediate Update:** For Identification (DINOv2), uploading images updates the Vector Database, instantly improving retrieval.
- **Periodic Retraining:** For Detail Detection (YOLO11), the collected feedback data is used to periodically retrain the model to resolve persistent confusion.

3.1.2 System Capabilities

Module	Capabilities
Personalized Tourist Experience	● Authentication: Secure login via Google/Email. ● User Profiling: Capture interest tags and behavior. ● Vectorization: Generate embeddings for user preferences. ● Smart Recommendations: Suggest places via similarity search.
Immersive AI Visual Discovery	● Image Processing: Capture and preprocess inputs. ● Landmark Recognition: Identify structures and architectural details. ● Context Retrieval: Fetch historical facts from the database. ● Narrative Generation: Create localized stories using LLM.
Intelligent Itinerary & Travel Companion	● Generative Planning: Create itineraries from text prompts. ● Constraint Validation: Check hours, location, and feasibility. ● Route Optimization: Optimize travel sequence and timing. ● Dynamic Adaptation: Adjust plans based on user vectors.
Smart Content Curation & Management	● Access Control: Role-based permissions (Admin/User). ● Content Ingestion: Create/update landmark details. ● Data Synchronization: Sync manual inputs with AI vectors. ● Validation: Human-in-the-loop accuracy monitoring.

Table 3: System Modules and Functional Capabilities

3.1.3 System Limitations

There are some limitations for this system as listed below:

1. Internet connectivity is required to connect the mobile app to cloud AI microservices (Vision Pipeline and LLM Agent).
2. Bad weather or low-light conditions can affect the accuracy of the landmark recognition.
3. The system can only recognize and explain landmarks that are pre-indexed in the system database.
4. Continuous use of the camera and GPS for discovery may result in high battery consumption.

3.2 Project Management

3.2.1 Work Breakdown Structure (WBS)

The project execution is structured into five main phases to ensure systematic development and deployment, as illustrated in Figure 11.

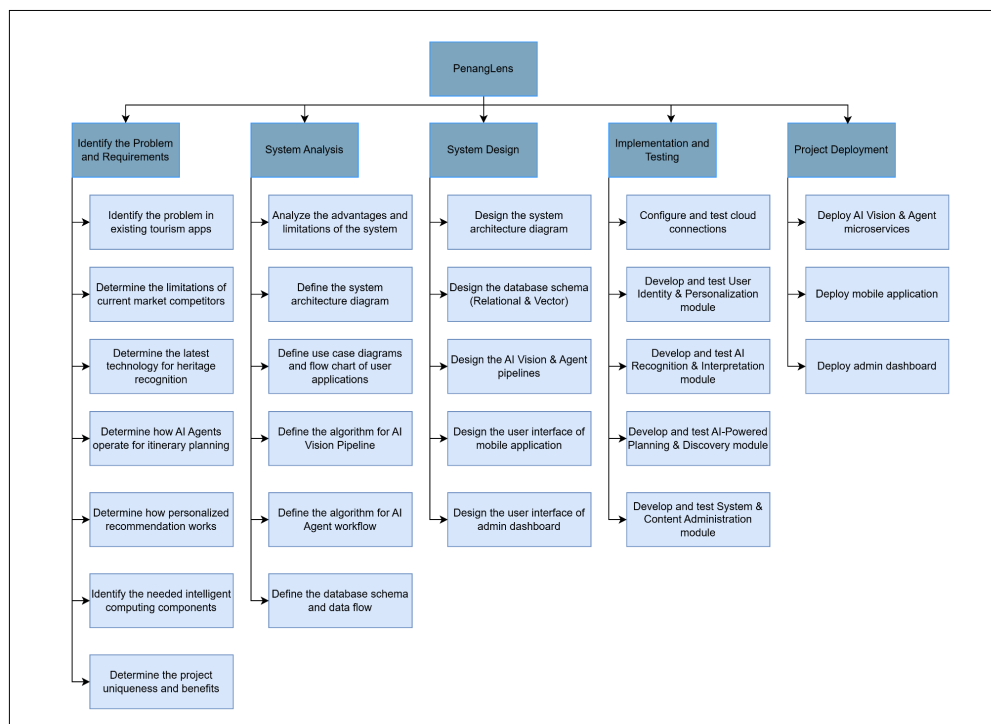


Figure 11 : Work Breakdown structure

3.2.2 Work Schedule (Gantt Chart & Milestone Timeline)

Milestone	Tasks	Deadline
1	Initial Proposal Submission	26 October 2025
2	System Requirement and Design (SRD) Report	15 December 2025
3	System Requirement and Design (SRD) Presentation	2 January 2026
4	Prototype Development and Testing	15 March 2026
5	Project Progress Demo	29 March 2026
6	Final System Development and Integration	17 May 2026
7	Final Report	24 May 2026
8	Final System Demo	7 June 2026

Table 4: Milestone of project

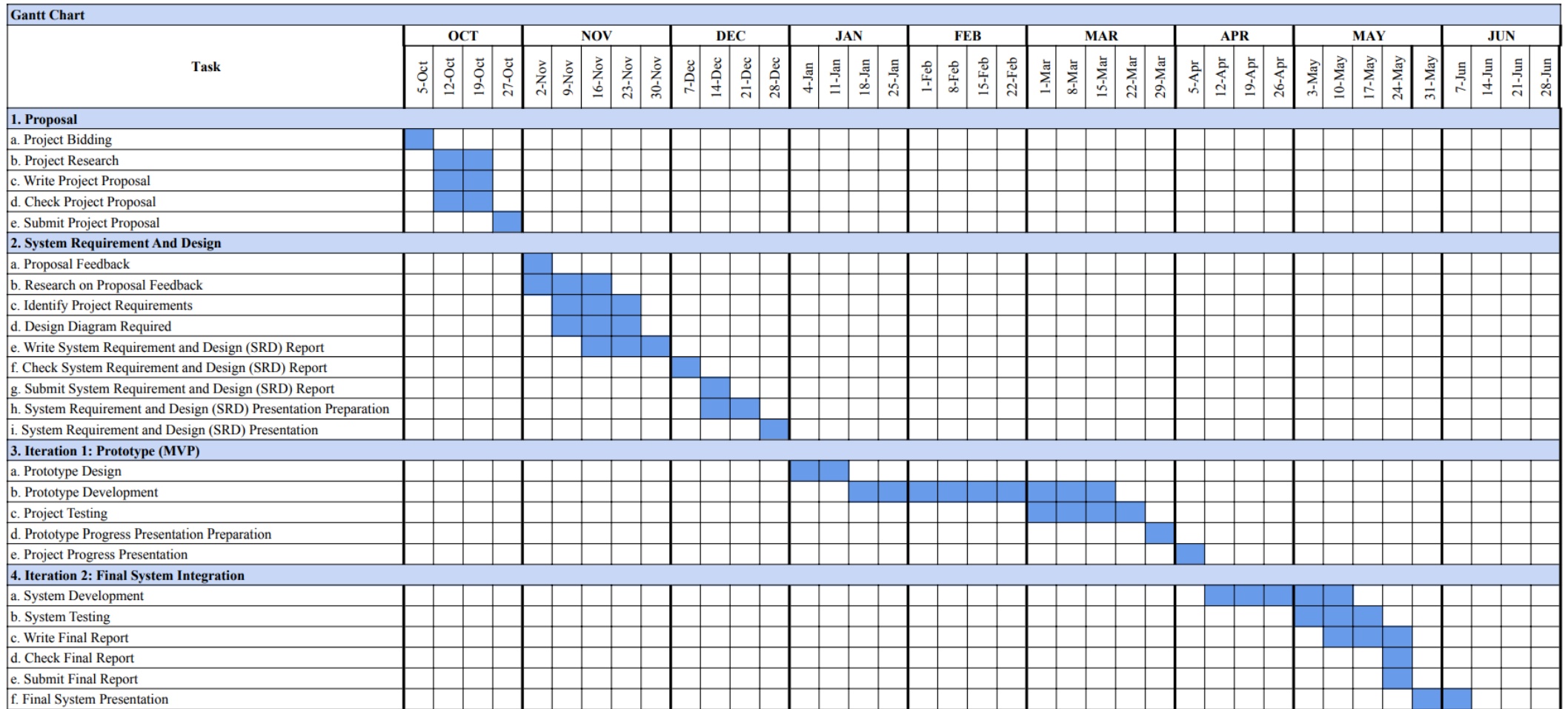


Figure 12 : Gantt Chart of PenangLens

3.2.3 SWOT Analysis

Table 5 summarizes the Strengths, Weaknesses, Opportunities, and Threats (SWOT) analysis of the PenangLens project.

Strengths	Weaknesses
1. Deep Visual Interpretation: Unlike generic tools, the system uses a unique "coarse-to-fine" AI pipeline (DINOv2 + YOLO11) to identify specific architectural details, not just general landmarks. 2. Proactive AI Agent: Features an autonomous agent (LangGraph) that actively validates and plans itineraries, moving beyond simple static maps. 3. Personalized Discovery: Recommendations are mathematically matched to user interest vectors (e.g., History vs. Nature), ensuring relevance. 4. Contextual Storytelling: Generates unique, localized stories using Generative AI (LLM) grounded in a verified database.	1. Dependency on Connectivity: The system requires a stable internet to access cloud-based AI microservices for recognition and planning. 2. High Resource Consumption: Continuous use of the camera and GPS for the Discovery Map will drain the user's mobile battery faster than standard apps. 3. Limited Knowledge Scope: The system can only recognize and explain landmarks that are explicitly indexed in the system's database.
Opportunities	Threats
1. Sustainable Tourism (SDG 11): The system can actively distribute tourist traffic to lesser-known heritage sites to reduce overcrowding at hotspots. 2. Scalability to Other Cities: The modular architecture allows the system to be easily adapted for other cities in Malaysia or even other countries once the solution is matured. 3. Data Insights: Aggregated anonymous data on tourist interests and routes can provide valuable insights for local tourism boards.	1. Rising Cloud Costs: High API costs for GPU processing (Vision AI) and LLM tokens could impact long-term operational sustainability. 2. Competitor Updates: Major tech platforms (e.g., Google) could release similar heritage-focused features, reducing the app's unique value. 3. AI Hallucinations: There is a risk of the LLM generating plausible but incorrect historical facts if the retrieval layer fails even if RAG has been implemented.

Table 5: SWOT Analysis of PenangLens

3.3 Development Methodology

This project adopts an **Iterative and Incremental Development Methodology**. While the project timeline follows a structured progression of milestones (as shown in the Gantt Chart in Figure 12), the technical development is executed in two distinct cycles: **Iteration 1** and **Iteration 2**.

Justification for Iterative Methodology:

1. **Managing AI Uncertainty:** AI components (Computer Vision and AI agents development) require continuous experimental tuning. A linear Waterfall model is unsuitable as it does not allow for revisiting the model parameters. The Iterative approach mitigates this risk by deploying a basic version of both the Vision Pipeline and the AI Agent in Iteration 1 to validate the core architecture, before optimizing accuracy and reasoning logic in Iteration 2.
2. **Structured Increments:** This methodology aligns with the project's academic milestones. The system is built incrementally, first as a functional Minimum Viable Product (MVP) covering the essential "happy paths" for the Prototype Presentation, and subsequently refined to handle complex edge cases and user feedback for the final submission.

Development Phases:

- **Iteration 1: Prototype (MVP) (Jan – March)**
 - Implement the basic end-to-end workflow within a local development environment. The Vision Module performs fundamental landmark identification and object detection, and the AI Agent handles simple narrative generation. The goal is to validate the core application logic and model performance before deployment.
- **Iteration 2: The Final System (April – May)**
 - Enhance the system based on feedback and execute cloud migration. The Vision Module is upgraded for more fine-grained details and the AI Agent gains complex reasoning , and all microservices are deployed to the Azure Cloud for the final production-ready system.

3.4 Requirement Elicitation Methodology

The requirements for the PenangLens system were derived through a Gap Analysis and Comparative Market Study and academic research, as detailed in **Section 2.3** and **Section 2.4** of this report. Instead of conducting user surveys, which rely on subjective self-reporting, this project adopted a data-driven approach by analyzing the functional limitations of existing market leaders (Google Lens, TripAdvisor, and Layla AI).

Consequently, the following key system capabilities were synthesized to directly resolve the critical market deficiencies identified as below:

- **Bridging the Semantic Gap:** To address the lack of historical context in current visual tools, the system requires context-aware educational stories that provide deep narrative interpretation rather than simple object labels.
- **Solving Information Overload:** To resolve the inefficiency of manual planning from scattered data, the requirements specify an Autonomous AI Agent to automate logistics via a unified chat interface.
- **Enabling Deep Personalization:** To counter the "one-size-fits-all" limitation, the system necessitates an interest-based recommendation engine using preference vectors for tailored suggestions.
- **Refining Visual Granularity:** To overcome generic labeling limitations, the system requires fine-grained detection to identify specific architectural details, bridging the gap between seeing and understanding.
- **Mitigating AI Reliability Risks:** To mitigate LLM hallucination risks, a "Human-in-the-Loop" monitoring system is established to enforce strict content verification.

3.5 Detailed Requirements of the New System

3.5.1 Functional Requirements

ID	Module	Requirement Description
FR-01	1	The system shall allow users to register and authenticate using third-party OAuth providers (Google Login) or email credentials.
FR-02	1	The system shall allow users to select personal interest tags and recommend relevant landmarks based on their personalization profile.
FR-03	2	The system shall allow users to capture or upload photos to identify landmarks and their specific architectural details.
FR-04	2	The system shall generate localized, context-aware educational stories about identified landmarks using AI.
FR-05	3	The system shall allow users to request personalized travel itineraries through a natural language chat interface.
FR-06	3	The system shall generate and display feasible travel routes on an interactive map, validating time and location constraints.
FR-07	3	The system shall allow users to interact with the map to view details of recommended Points of Interest (POIs).
FR-08	4	The system shall allow administrators to manage the hierarchical content of landmarks and sub-POIs.
FR-09	4	The system shall provide a dashboard for administrators to monitor the performance and accuracy of the AI models.

Table 6: Functional Requirements Specification of PenangLens

3.5.2 Non-Functional Requirements

1. System Reliability

- The system shall perform all required functionalities reliably, provided a stable internet connection (Mobile Data or Wi-Fi) is maintained.
- The failure in a system module should not affect the other module's functionalities.

2. System Security

- The system will perform secure access control for user accounts and the admin dashboard through third-party authentication (Clerk) and role-based authorization.

3. System Performance

- The system shall offload computationally intensive AI tasks (Vision/LLM) to cloud microservices to ensure the mobile client remains lightweight and responsive.
- The system shall be able to recognize landmarks and return details within 5 to 8 seconds under standard network conditions.

4. System Scalability

- The backend architecture shall be capable of scaling AI services independently to handle spikes in user traffic without performance degradation.

5. Data Integrity

- The system shall enforce a Human-in-the-Loop (HITL) verification step for all historical facts before they are indexed into the vector database to prevent AI hallucinations.

3.6 Analysis of New System (UML Modeling)

3.6.1 Use Case Diagram

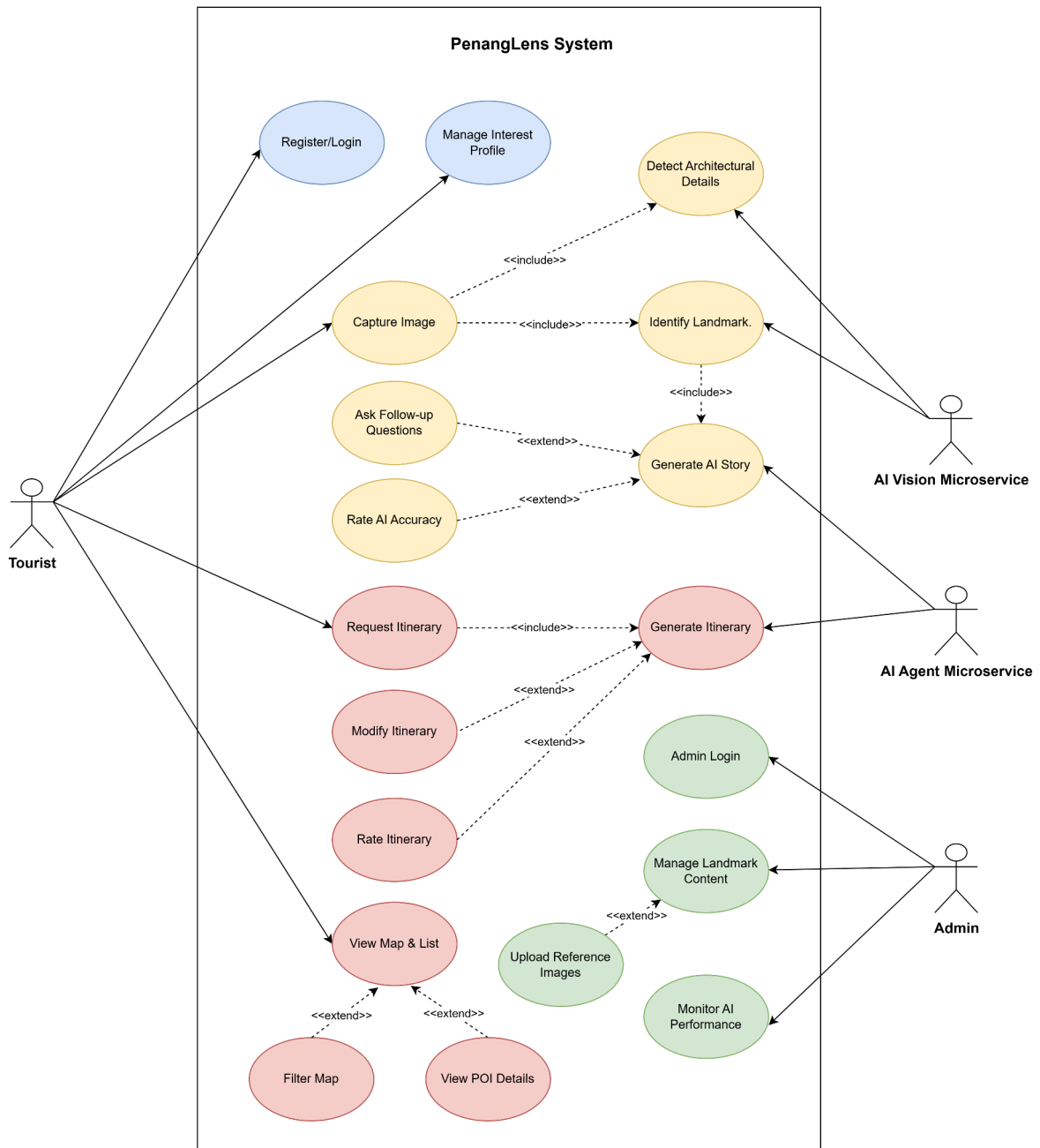


Figure 13 : Overall Use case diagram of Penang Lens

ID	Use Case Name	Description
UC-01	Register / Login	Allows users to authenticate using OAuth (Google) or email credentials.
UC-02	Manage Interest Profile	Allows users to select interest tags to create a preference vector for tailored recommendations.
UC-03	Capture Image	Users capture a photo with the in-app camera to begin the recognition process.
UC-04	Identify Landmark	The DINOv2 model performs a k-NN search to retrieve the landmark ID from the vector database.
UC-05	Detect Architectural Details	The YOLO11 model detects specific fine-grained features within the image.
UC-06	Generate AI Story	The Gemini LLM uses Retrieval-Augmented Generation (RAG) to craft a context-aware narrative.
UC-07	Ask Follow-up Questions	Allows the user to engage in a dialogue to probe deeper into topics related to the current landmark.
UC-08	Rate AI Accuracy	Enables users to provide feedback on the landmark identification for model training.
UC-09	Request Itinerary	Users request a travel plan using complex logistical queries in natural language.
UC-10	Generate Itinerary	The LangGraph-based agent reasons and acts (ReAct) to produce a feasible, customized travel plan.
UC-11	View Map & List	Displays recommended or planned POIs on an interactive map using the user's location.
UC-12	Filter Map	Allows users to narrow down the displayed POIs by category or proximity.
UC-13	View POI Details	Displays metadata like opening hours, ticket prices, and historical summaries for selected locations.
UC-14	Modify Itinerary	Users request changes to the existing itinerary, triggering the agent to re-run its planning logic.
UC-15	Rate Itinerary	Allows the user to rate the quality of the generated travel plan.

UC-16	Admin Login	Enables secure authentication for administrators to access the web dashboard.
UC-17	Manage Landmark Content	Allows admins to update the text descriptions and hierarchical structure of heritage data.
UC-18	Upload Reference Images	Enables admins to upload photos for automated DINOv2 feature vector extraction and database indexing.
UC-19	Monitor AI Performance	Provides a dashboard to visualize model accuracy, user feedback, and system usage (e.g., Confusion Matrix).

Table 7: Use Case List of PenangLens

3.6.2 Use Case Descriptions

Module 1: Personalized Tourist Experience

Use case ID	UC-01
Use case name	Register / Login
Brief Description	Users create a new account or log into an existing account to access personalized features and save history.
Triggering event	The user opens the application and selects "Sign In" or "Register".
Actor	Tourist
Related use case	-
Pre-conditions	Internet connection is available.
Post-conditions	The user is authenticated and redirected to the home screen.
Flows of activity	Actor 1. The user selects the login method (Email or Google). 2. The user enters credentials (if Email) or confirms Google account. System 3. System validates credentials via Identity Provider. 4. The system generates session tokens. 5. The system retrieves user profile data from the database. 6. The system redirects the user to the Home Screen.
Exception conditions	1. Invalid username or password. 2. Network timeout during authentication. 3. User account is suspended.

Table 8: Use Case Specification – Register / Login (UC-01)

Use case ID	UC-02
Use case name	Manage Interest Profile
Brief Description	Users select interest tags (e.g., History, Street Art) to allow the system to build a preference vector for personalized recommendations.
Triggering event	The user navigates to the "Profile" tab or completes first-time onboarding.
Actor	Tourist
Related use case	-
Pre-conditions	The user is logged in (UC-01).
Post-conditions	The user preference vector is updated in the database.
Flows of activity	Actor 1. User views list of interest categories. 2. The user selects or deselects specific tags (e.g., "Heritage", "Food"). 3. User taps "Save Profile". System 4. The system calculates new preference vectors based on tags. 5. System saves updated profile to database. 6. System displays "Preferences Saved" confirmation.
Exception conditions	1. Database connection failure. 2. The user attempts to save an empty profile (if minimum selection required).

Table 9: Use Case Specification – Manage Interest Profile (UC-02)

Module 2: Immersive AI Visual Discovery

Use case ID	UC-03
Use case name	Capture Image
Brief Description	Users capture a photo of a landmark using the in-app camera to initiate the recognition process.
Triggering event	The user points the camera at a building and presses the shutter button.
Actor	Tourist
Related use case	Includes UC-04 Identify Landmark Includes UC-05 Detect Architectural Details
Pre-conditions	Camera permission is granted.
Post-conditions	The image is captured and sent for processing.
Flows of activity	Actor 1. The user aligns landmarks within the camera frame. 2. The user taps the capture button. System 3. The system freezes the image frame. 4. The system compresses images for optimization. 5. The system displays "Analyzing..." loader. 6. System initiates parallel processing for UC-04 Identify Landmark and UC-05 Detect Architectural Details.
Exception conditions	1. Camera hardware failure. 2. Storage full (cannot save temporary image).

Table 10 : Use Case Specification – Capture Image (UC-03)

Use case ID	UC-04
Use case name	Identify Landmark
Brief Description	The system analyzes the captured image using the AI Vision Microservice (DINOv2) to retrieve the specific landmark identity from the vector database.
Triggering event	Image is captured from UC-03 Capture Image.
Actor	AI Vision Microservice
Related use case	Included by UC-03 Capture Image Includes UC-06 Generate AI Story
Pre-conditions	Image data is valid.
Post-conditions	Landmark ID is identified and architectural details are detected.
Flows of activity	1. System extracts feature vector from image (DINOv2). 2. The system performs k-NN search in the Vector Database. 3. The system calculates similarity scores for top matches. 4. System retrieves the top matching Landmark ID with a confidence score > 85% 5. The system triggers UC-06 to generate a story.
Exception conditions	1. No matching landmark found (low similarity score). 2. The image is too blurry for analysis. 3. Cloud API timeout.

Table 11 : Use Case Specification – Identify Landmark (UC-04)

Use case ID	UC-05
Use case name	Detect Architectural Details
Brief Description	The system detects specific features (e.g., "Carving," "Tiles") within the image to add granular context to the analysis.
Triggering event	Image is captured from UC-03 Capture Image.
Actor	AI Vision Microservice
Related use case	Included by UC-03 Capture Image
Pre-conditions	Valid image data is available.
Post-conditions	The processed image with labeled bounding boxes capturing the details is returned.
Flows of activity	1. The system runs inference on the image using the fine-tuned YOLO11 model. 2. The system retrieves the Landmark ID (from completed UC-04). 3. The system retrieves the specific class list (e.g., "Roofs", "Carvings") associated with that Landmark ID. 4. The system filters raw detections against the specific class list to remove false positives. 5. The system returns the image with labeled bounding boxes capturing the details.
Exception conditions	1. No specific details were detected (only the main building found).

Table 12 : Use Case Specification – Detect Architectural Details (UC-05)

Use case ID	UC-06
Use case name	Generate AI Story
Brief Description	The AI Agent Microservice generates a localized, fact-checked narrative about the landmark using the identified ID and details.
Triggering event	Completion of analysis data from UC-04 Identify Landmark and UC-05 Detect Architectural Details.
Actor	AI Agent Microservice
Related use case	Included by UC-04 Identify Landmark Extended by UC-07 Ask Follow-up Questions Extended by UC-08 Rate AI Accuracy
Pre-conditions	Landmark ID and details are available.
Post-conditions	Story text is displayed to the user.
Flows of activity	1. The system retrieves historical facts for Landmark ID 2. The system combines historical facts and detected visual details (from UC-05) into a prompt.. 3. LLM (Gemini) generates the narrative. 4. The system returns the story text to the user interface.
Exception conditions	1. LLM service unavailable. 2. Content generation blocked by safety filters.

Table 13 : Use Case Specification – Generate AI Story (UC-06)

Use case ID	UC-07
Use case name	Ask Follow-up Questions
Brief Description	The user engages in a chat dialogue with the AI to learn more about the specific landmark presented in the story.
Triggering event	The user types a question in the chat box after story generation.
Actor	Tourist, AI Agent Microservice
Related use case	Extends UC-06 Generate AI Story
Pre-conditions	A story context UC-06 Generate AI Story is active.
Post-conditions	AI provides a specific answer to the user's query.
Flows of activity	Actor 1. The user submits a text query (e.g., "When was this built?"). System 2. The system retrieves relevant info from the vector store. 3. AI generates answers, maintaining context of the current landmark. 4. The system displays the answer in the chat interface.
Exception conditions	1. The query is unrelated to the landmark (AI prompts users to stay on topic). 2. Network failure.

Table 14 : Use Case Specification – Ask Follow-up Questions (UC-07)

Use case ID	UC-08
Use case name	Rate AI Accuracy
Brief Description	The user provides feedback (Thumbs Up/Down) on whether the landmark identification was correct.
Triggering event	The user taps the rating icon on the result screen.
Actor	Tourist
Related use case	Extends UC-06 Generate AI Story
Pre-conditions	AI Result is displayed.
Post-conditions	Feedback data is saved for model retraining.
Flows of activity	Actor 1. The user evaluates the result. 2. User taps rating button (Correct/Incorrect). System 3. The system logs the image ID and rating to the Feedback Database. 4. The system displays a "Thanks for feedback" toast message.
Exception conditions	1. Server fails to acknowledge feedback.

Table 15 : Use Case Specification – Rate AI Accuracy (UC-08)

Module 3: Intelligent Itinerary & Travel Companion

Use case ID	UC-09
Use case name	Request Itinerary
Brief Description	The user requests a travel plan using natural language (e.g., "Plan a 3-hour heritage walk").
Triggering event	User types prompt in the "Planner" tab.
Actor	Tourist
Related use case	Includes UC-10 Generate Itinerary
Pre-conditions	The user is logged in.
Post-conditions	The request is parsed and sent to the agent.
Flows of activity	Actor 1. The user enters travel constraints (Time, Interest, Location). 2. The user taps "Generate Plan". System 3. The system validates input length. 4. System initiates UC-10 Generate Itinerary.
Exception conditions	1. Input is nonsensical.

Table 16 : Use Case Specification – Request Itinerary (UC-09)

Use case ID	UC-10
Use case name	Generate Itinerary
Brief Description	The AI Agent plans a route by validating constraints, checking opening hours, and calculating travel times.
Triggering event	Request received from UC-09 Request Itinerary.
Actor	AI Agent Microservice
Related use case	Included by UC-09 Request Itinerary Extended by UC-14 Modify Itinerary, UC-15 Rate Itinerary
Pre-conditions	User preference vector is available.
Post-conditions	A feasible itinerary is displayed.
Flows of activity	1. Agent identifies potential spots matching user interest. 2. The agent checks opening hours for requested time. 3. The agent calculates travel distance between spots. 4. The agent optimizes the sequence. 5. System returns structured itinerary list.
Exception conditions	1. No locations found matching criteria. 2. Impossible constraints (e.g., "Visit 10 museums in 1 hour").

Table 17 : Use Case Specification – Generate Itinerary (UC-10)

Use case ID	UC-11
Use case name	View Map & List
Brief Description	User views recommended places or the generated itinerary on an interactive map.
Triggering event	The user opens the "Explore" tab or Itinerary result.
Actor	Tourist
Related use case	Extended by UC-12 Filter Map Extended by UC-13 View POI Details
Pre-conditions	GPS location is enabled.
Post-conditions	Map populated with POI pins.
Flows of activity	System 1. System detects user GPS location. 2. The system retrieves relevant POIs from the database. 3. System renders map with markers. Actor 4. The user pans or zooms the map to explore.
Exception conditions	1. GPS signal lost. 2. Maps API quota exceeded.

Table 18 : Use Case Specification – View Map & List (UC-11)

Use case ID	UC-12
Use case name	Filter Map
Brief Description	The user applies filters (Category, Radius) to narrow down the displayed Points of Interest.
Triggering event	The user taps the "Filter" button on the map.
Actor	Tourist
Related use case	Extends UC-11 View Map & List
Pre-conditions	Map is active.
Post-conditions	Map view updates with filtered results.
Flows of activity	Actor 1. The user selects filter criteria (e.g., "Food only", "Within 1km"). 2. The user taps "Apply". System 3. System queries database with new constraints. 4. System refreshes map markers.
Exception conditions	1. No results match the filter.

Table 19 : Use Case Specification – Filter Map (UC-12)

Use case ID	UC-13
Use case name	View POI Details
Brief Description	The user selects a specific pin to view detailed information like opening hours and ticket prices.
Triggering event	The user taps a marker on the map.
Actor	Tourist
Related use case	Extends UC-11 View Map & List
Pre-conditions	POI data is available.
Post-conditions	Detail modal/screen is displayed.
Flows of activity	Actor 1. The user clicks a specific map marker. System 2. The system fetches detailed metadata for POI. 3. The system displays info cards (Name, Rating, Hours).
Exception conditions	1. Data load failure.

Table 20 : Use Case Specification – View POI Details (UC-13)

Use case ID	UC-14
Use case name	Modify Itinerary
Brief Description	The user requests a change to the generated plan (e.g., "Remove the third stop"), triggering a re-generation.
Triggering event	The user selects "Remake" or types modification request.
Actor	Tourist
Related use case	Extends UC-10 Generate Itinerary
Pre-conditions	An itinerary exists.
Post-conditions	Updated itinerary is displayed.
Flows of activity	Actor 1. The user taps "Remake Plan" or prompting for change to the existing itinerary. System 3. System re-runs logic UC-10 Generate Itinerary. 4. The system presents a new optimized plan.
Exception conditions	1. Modification makes plans impossible.

Table 21 : Use Case Specification – Modify Itinerary (UC-14)

Use case ID	UC-15
Use case name	Rate Itinerary
Brief Description	The user rates the quality of the suggested travel plan.
Triggering event	User taps rating stars on itinerary view.
Actor	Tourist
Related use case	Extends UC-10 Generate Itinerary
Pre-conditions	Itinerary is generated.
Post-conditions	Rating saved.
Flows of activity	Actor 1. The user taps a star rating (1-5). System 2. System saves rating to database. 3. System adjusts user preference weights for future recommendations. 4. The system displays a "Thank you" message.
Exception conditions	1. If network connection is unavailable, the system displays a 'Network Error' toast and discards the rating. 2. If the backend fails to respond, the system displays a "Failed to submit rating" error message and allows the user to try again.

Table 22 : Use Case Specification – Rate Itinerary (UC-15)

Module 4: Smart Content Curation & Management

Use case ID	UC-16
Use case name	Admin Login
Brief Description	Administrator authenticates securely to access the web dashboard.
Triggering event	Admin navigates to the web portal URL.
Actor	Admin
Related use case	-
Pre-conditions	Admin account exists in the database.
Post-conditions	Admin Dashboard is displayed.
Flows of activity	Actor 1. Admin enters secure credentials. System 2. System validates role authorization. 3. The system grants access to the dashboard.
Exception conditions	1. Unauthorized access attempt.

Table 23 : Use Case Specification – Admin Login (UC-16)

Use case ID	UC-17
Use case name	Manage Landmark Content
Brief Description	Admin adds, edits, or deletes landmark information and historical facts.
Triggering event	Admin selects the "Content Management" menu.
Actor	Admin
Related use case	Extended by UC-18 Upload Reference Images
Pre-conditions	Admin is logged in.
Post-conditions	Landmark data is updated in the database.
Flows of activity	Actor 1. Admin searches for landmarks. 2. Admin edits text fields (Description, History). 3. Admin clicks "Save". System 4. System updates Relational DB. 5. System triggers re-indexing of embeddings.
Exception conditions	1. Data validation error (missing required fields).

Table 24 : Use Case Specification – Manage Landmark Content (UC-17)

Use case ID	UC-18
Use case name	Upload Reference Images
Brief Description	Admin uploads verified photos of landmarks to index them into the AI Vision microservice database.
Triggering event	Admin clicks "Add Photos" inside UC-17 Manage Landmark Content.
Actor	Admin, AI Vision Microservice
Related use case	Extends UC-17 Manage Landmark Content
Pre-conditions	Valid image file format.
Post-conditions	Image feature vector is added to Vector Database.
Flows of activity	Actor 1. Admin selects image files from local drive. 2. Admin clicks "Upload". System 3. The system uploads files to storage. 4. AI Vision Service extracts vectors (DINOv2). 5. The system saves vectors to index.
Exception conditions	1. File format not supported. 2. Vector extraction failure.

Table 25 : Use Case Specification – Upload Reference Images (UC-18)

Use case ID	UC-19
Use case name	Monitor AI Performance
Brief Description	Admin views analytics regarding model accuracy, user ratings, and system usage.
Triggering event	Admin selects the "Analytics" menu.
Actor	Admin
Related use case	-
Pre-conditions	Usage data is available.
Post-conditions	Analytics charts are displayed.
Flows of activity	Actor - Admin selects date range. System - The system aggregates feedback data (Confusion Matrix). - The system renders performance graphs and charts.
Exception conditions	No data available for selected range.

Table 26 : Use Case Specification – Monitor AI Performance (UC-19)

3.6.3 System Sequence Diagrams (SSD)

SSD 1: AI Vision Pipeline (Identify Landmark)

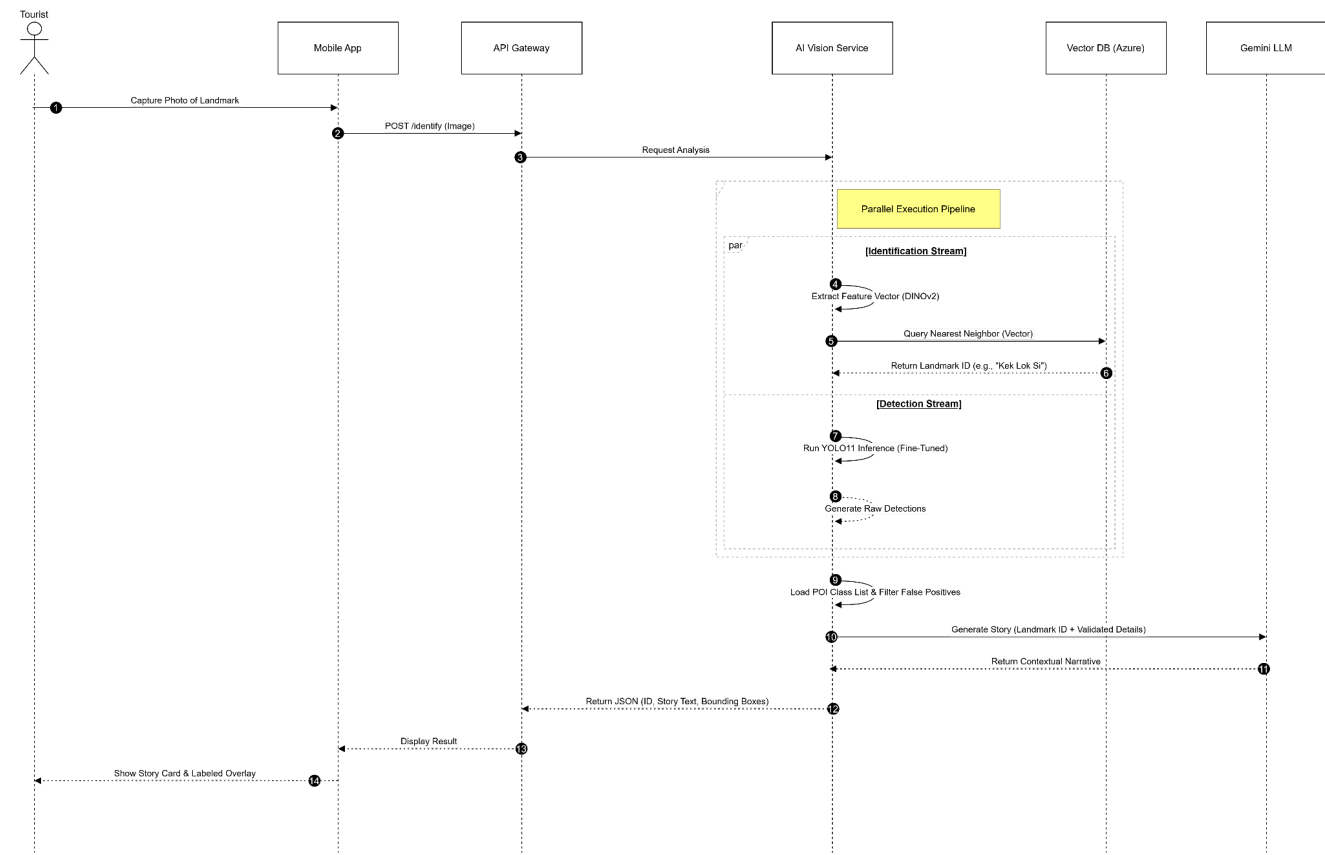


Figure 14 : System Sequence Diagram for AI Vision Pipeline

This sequence diagram in Figure 14 captures the core functionality of the **AI Recognition & Interpretation Module**. The process ensures that computationally intensive tasks are offloaded to the cloud.

The sequence begins with the Tourist taking a photograph. The AI Vision Service executes a parallel processing pipeline: one stream performs feature extraction using DINOv2 to query the Vector Database for the Landmark ID, while the second stream concurrently runs YOLO11 inference to detect fine-grained architectural details.

Once both streams complete, the system performs a filtering step, validating the raw YOLO11 detections against the specific Class List associated with the identified Landmark.

Finally, the validated data is sent to the Gemini LLM to generate the context-aware story. The workflow concludes when the system returns a unified JSON payload containing the narrative, landmark ID, and bounding box coordinates. The Mobile App uses this data to render and draw the bounding boxes around the architectural details directly on the user's screen alongside the story card.

SSD 2: AI Agent Workflow (Generate Itinerary)

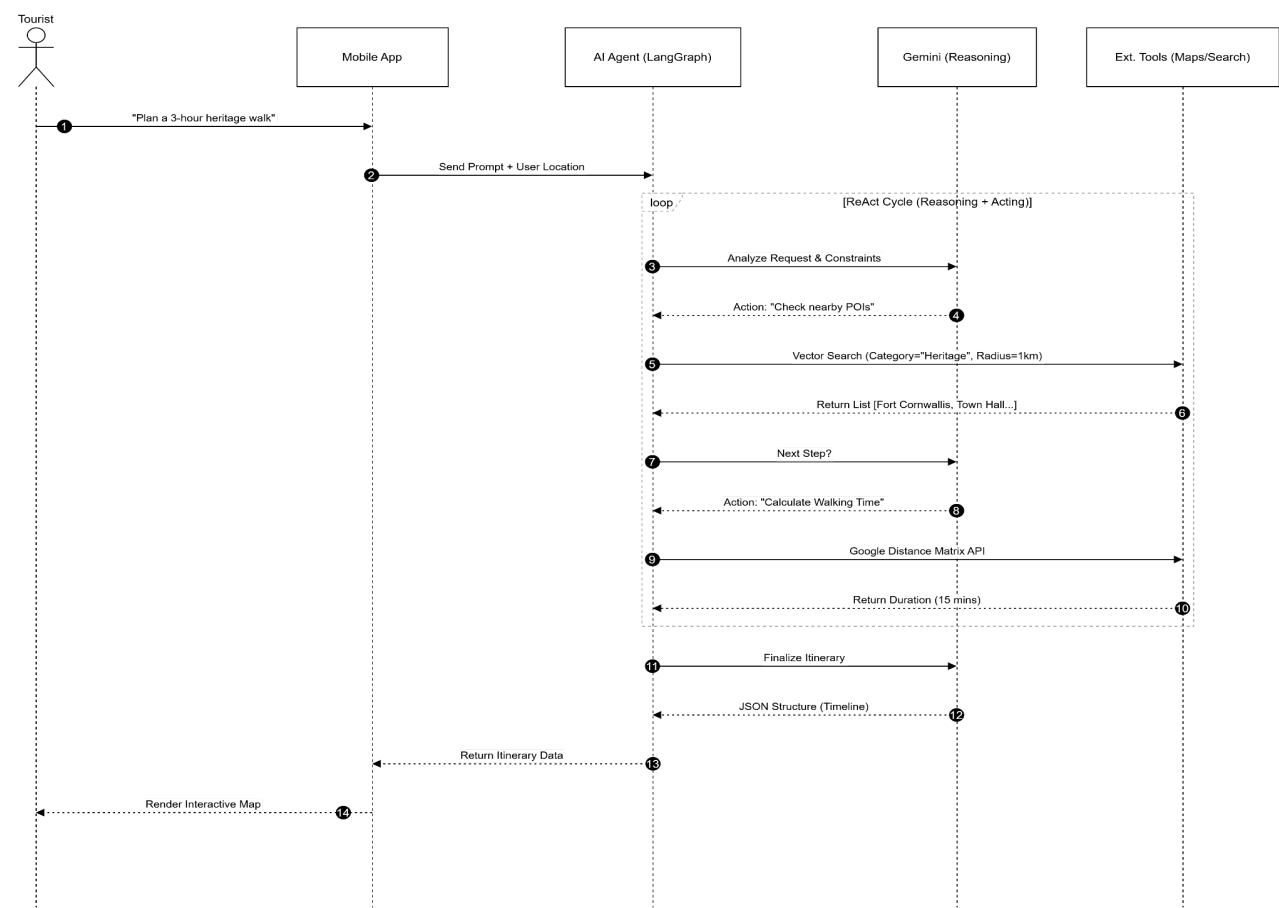


Figure 15 : System Sequence Diagram for AI Agent Itinerary Planning

This sequence details the proactive planning process managed by the **Intelligent Itinerary & Travel Companion Module (Module 3)**. It highlights the ReAct (Reasoning + Acting) paradigm implemented using LangGraph.

When the Tourist submits a natural language request, the AI Agent initiates a loop where the Gemini LLM reasons about the required actions (Tools). The agent calls External Tools (like a Distance Matrix or Vector Search) to gather real-time data. This gathered data is used by the LLM for subsequent reasoning steps, allowing the Agent to dynamically assemble and finalize a logical, feasible itinerary before sending the structured JSON back to the Mobile App for rendering.

Finally, the application parses this data to render an interactive map, plotting the optimized route and stops directly on the user's mobile interface for immediate exploration.

SSD 3: Admin Content Curation (Upload & Index)

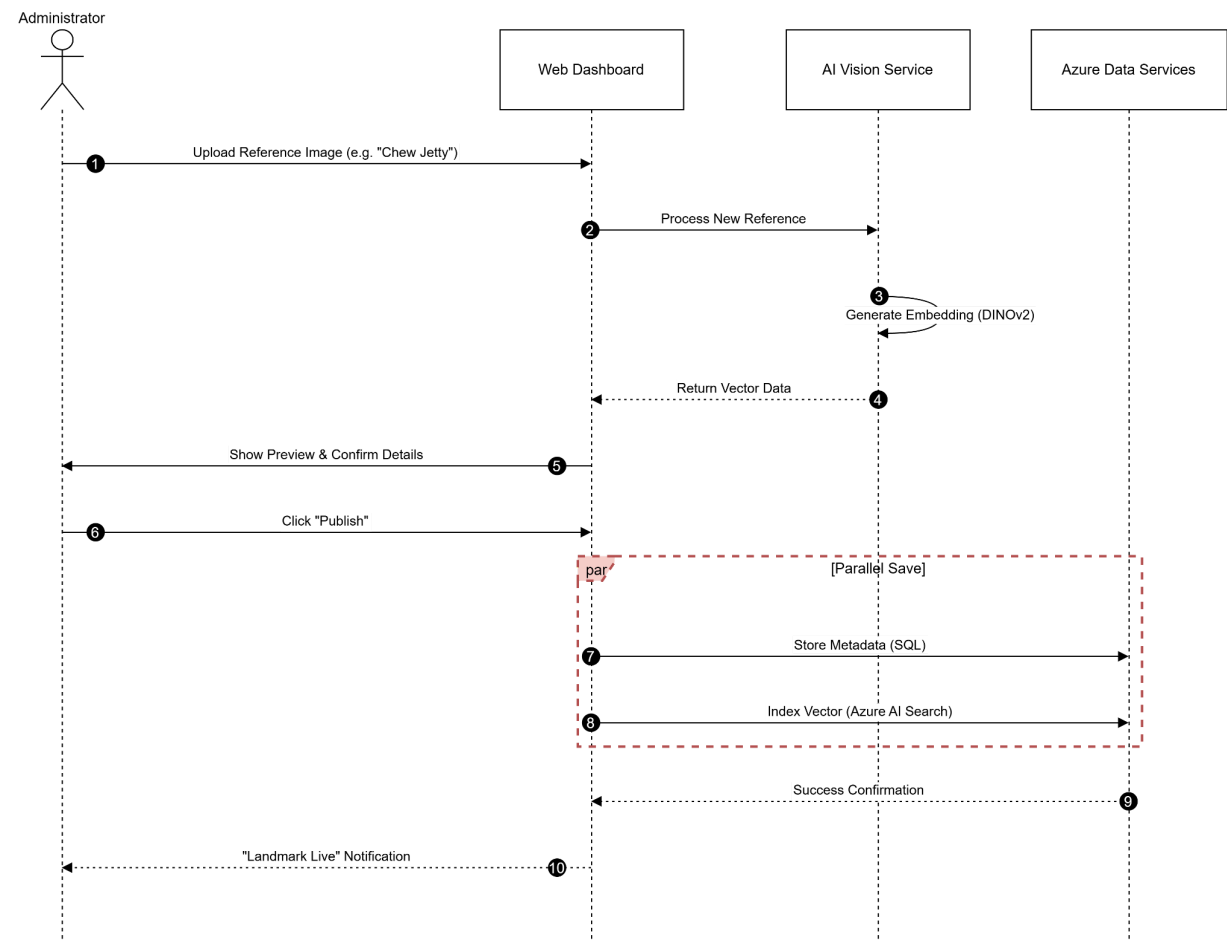


Figure 16 : Sequence Diagram for Human-in-the-Loop Data Ingestion

This sequence illustrates the *Human-in-the-Loop* (HITL) architecture for managing the database for the **Smart Content Curation & Management Module**.

The process, initiated by the **Administrator**, ensures that all data supporting the AI is verified and indexed correctly. When an image is uploaded, the **AI Vision Service** automatically extracts its feature vector using DINOv2 for immediate indexing into the **Vector DB**. This decoupling ensures that high-quality, AI-ready assets are generated without manual intervention, supporting the scalability of the system. The process concludes with the update of both the relational metadata (SQL) and the vector index.

Finally, the system returns a success confirmation to the dashboard and triggers a 'Landmark Live' notification to the Administrator, verifying that the new content is fully active and searchable by tourists.

3.6.4 State Diagram

3.6.4.1 AI Recognition & Interpretation Pipeline

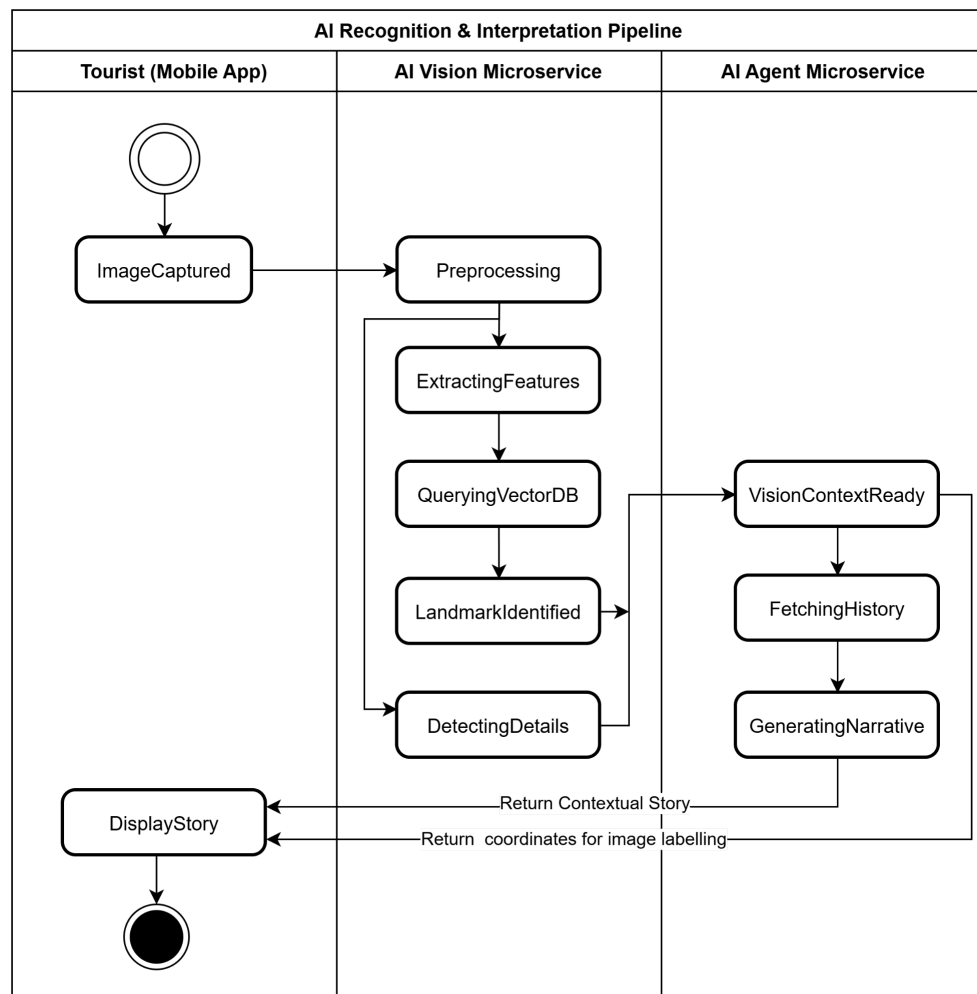


Figure 17 : State Diagram for AI Recognition & Interpretation Pipeline

Figure 17 illustrates the "Coarse-to-Fine" analysis pipeline. Initiated by the Tourist capturing an image, the AI Vision Microservice executes a parallel workflow:

Stream 1: Performs **ExtractingFeatures** (DINOv2) to reach **LandmarkIdentified**.

Stream 2: Concurrently executes **DetectingDetails** using YOLO11.

Synchronizing at **VisionContextReady**, the flow branches to immediately return a JSON payload for image labelling while concurrently triggering the AI Agent for **GeneratingNarrative**. These outputs converge at **DisplayStory**, where the mobile app integrates the visual overlay with the generated context for the user.

3.6.4.2 AI Agent Itinerary Planning Workflow

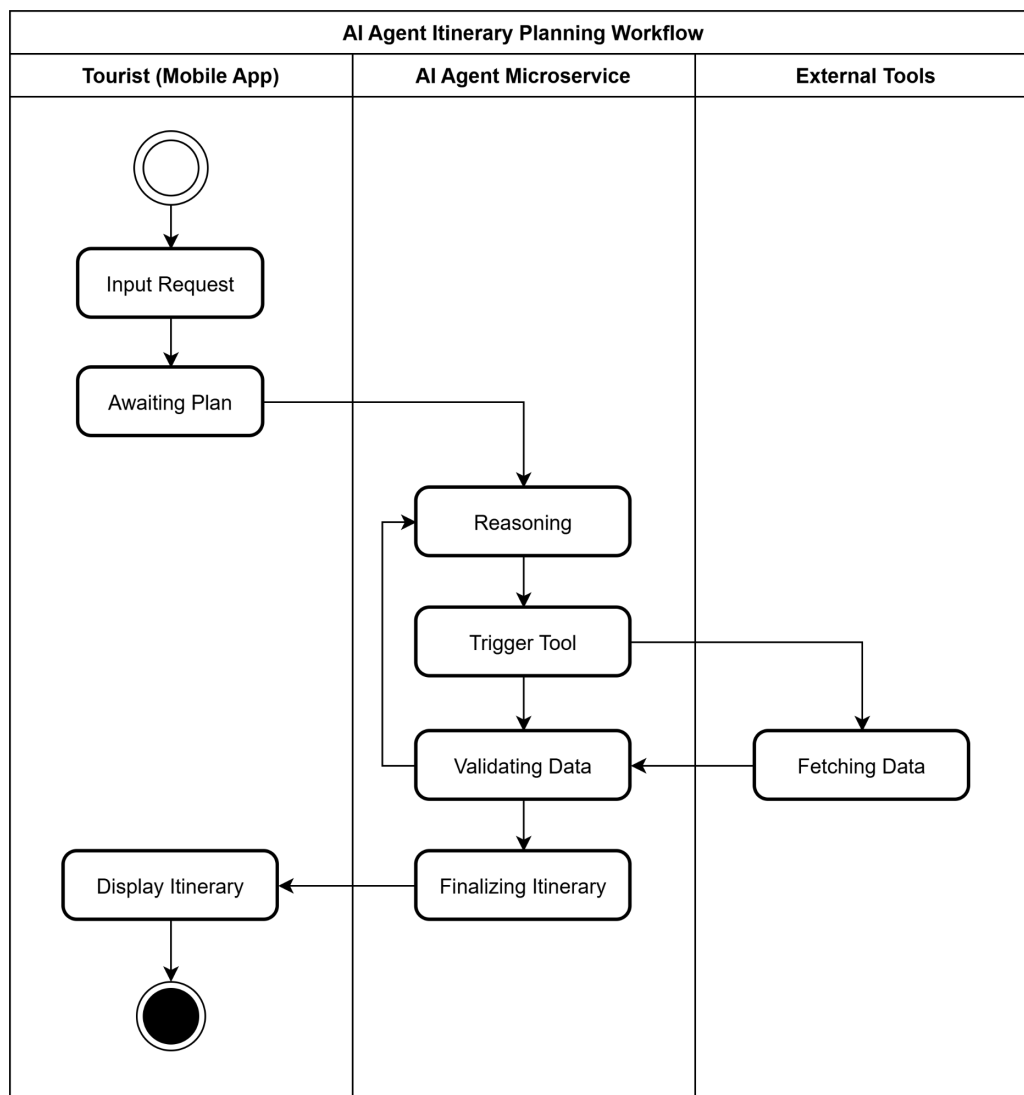


Figure 18 : State Diagram for AI Agent Itinerary Planning Workflow

Figure 18 models the proactive reasoning of the AI Agent. The flow begins with the **Tourist** sending a natural language request, triggering the **AI Agent Microservice**. The agent enters a **ReAct (Reason + Act)** cognitive loop: reasoning, triggering **External Tools** (Maps/Vector DB) for data, and validating the results.

A critical feature is the **Self-Correction Loop** where if the fetched data violates user constraints (e.g., closed venues), the agent transitions back to the *Reasoning* state to adjust the plan. Only when constraints are met does the system transition to *Finalizing Itinerary*.

3.6.4.3 AI-Assisted Content Curation Workflow

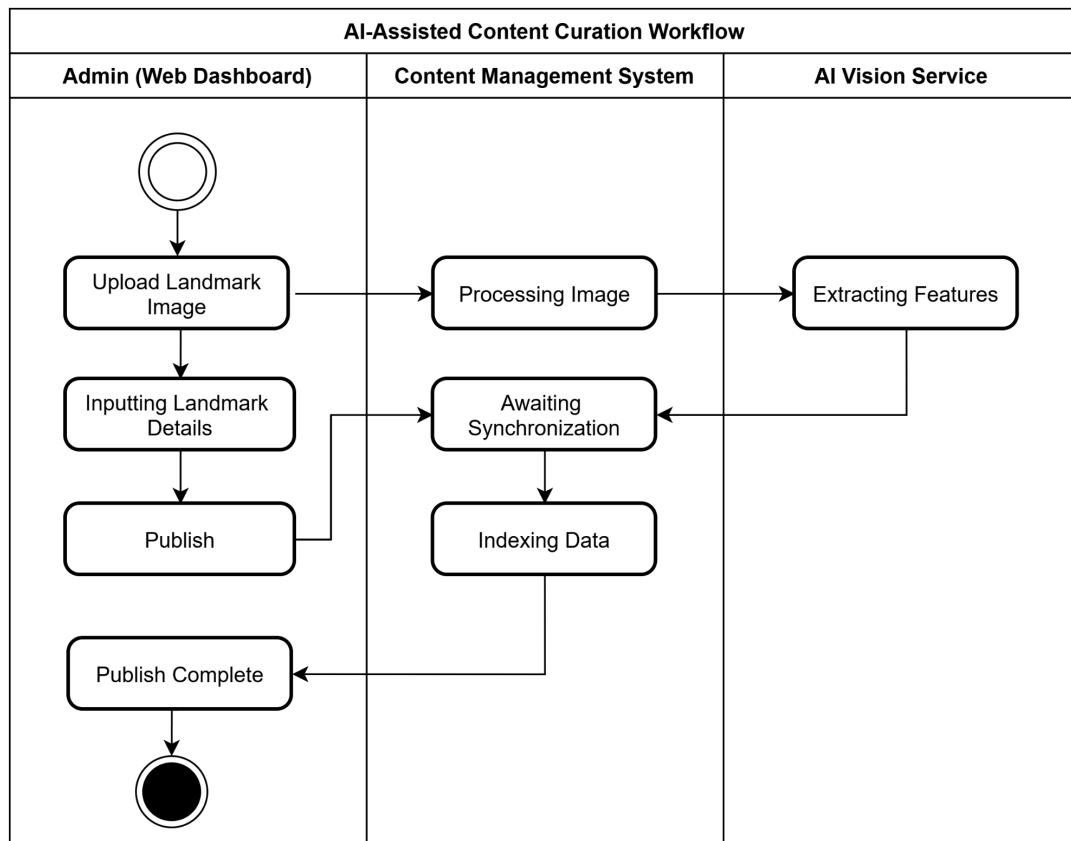


Figure 19 : State Diagram for AI-Assisted Content Curation Workflow

Figure 19 illustrates the "Human-in-the-Loop" data ingestion pipeline. To solve the "Cold Start" problem, the system uses parallel processing:

1. **Admin:** Manually transitions to *Inputting Landmark Details* (history, opening hours).
2. **AI Vision Service:** Concurrently enters the *Extracting Features* state in the background to generate DINOv2 vector embeddings.

The **Content Management System (CMS)** acts as the synchronizer in the *Awaiting Synchronization* state. It holds the process until it receives both the text from the Admin and the vector from the AI, ensuring complete data indexing before confirming success .

3.6.5 Entity Relationship Diagram (ERD)

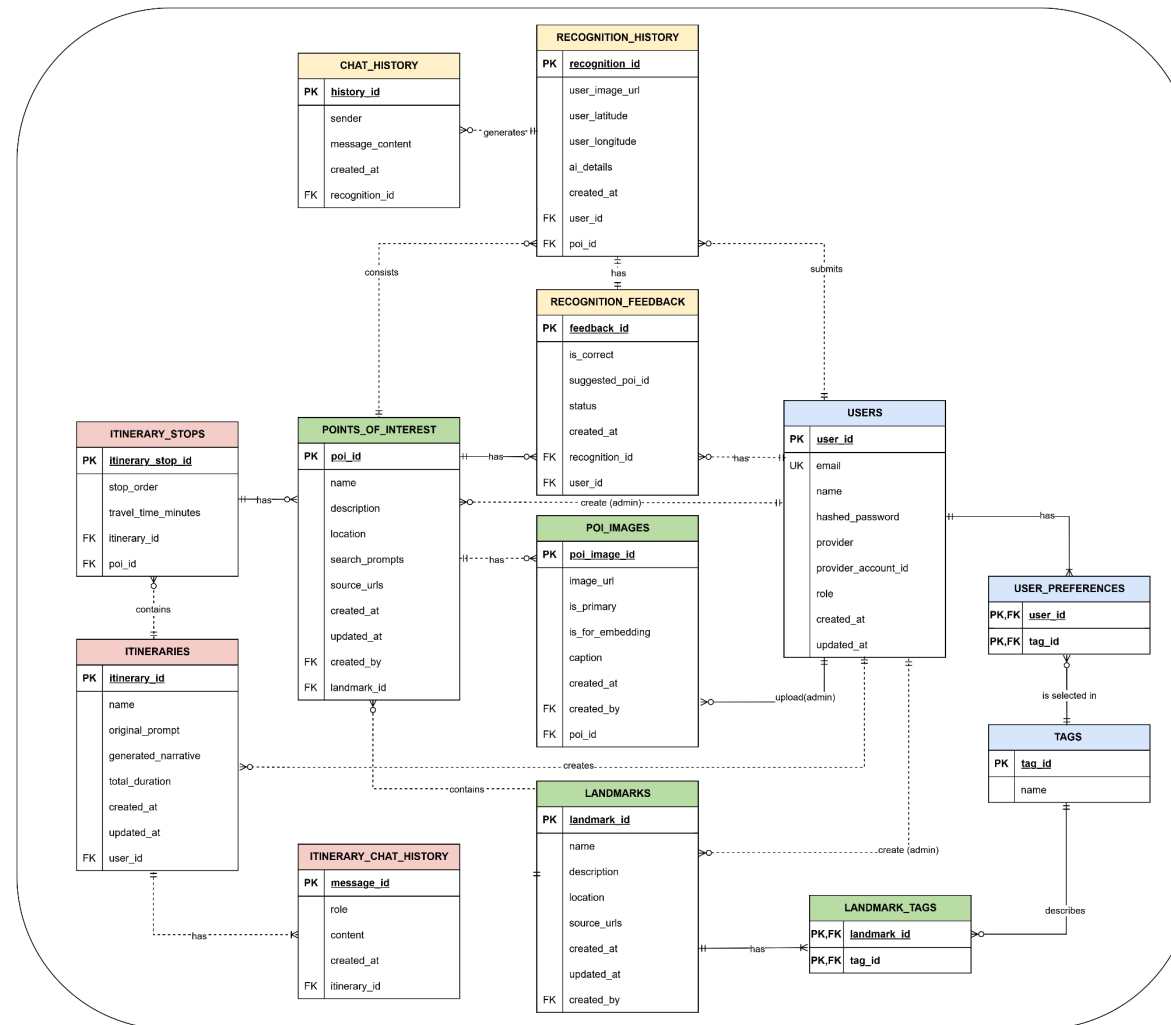


Figure 20 : Entity Relation Diagram of Penang Lens

. Key structural elements in Figure 20 include:

- **Hierarchical Content:** A one-to-many relationship between *Landmarks* and *Points_of_interest* supports the system's fine-grained architectural recognition.
- **AI Performance:** The *Recognition_history* and *Recognition_feedback* tables function as a system log to track model accuracy and user corrections for the Admin Dashboard.
- **Agent Planning:** *Itineraries* and *Itinerary_stops* persist the multi-step travel plans generated by the AI Agent.
- **Personalization:** *User_preferences* links users to interest *Tags*, enabling the semantic recommendation engine.

3.7 Technology Deployed

The proposed system is built on a scalable microservices architecture hosted on Microsoft Azure. This design decouples the core functionalities, allowing the AI-intensive tasks to be managed independently from the main application logic, ensuring high performance and reliability. The diagram below illustrates the key components and their interactions.

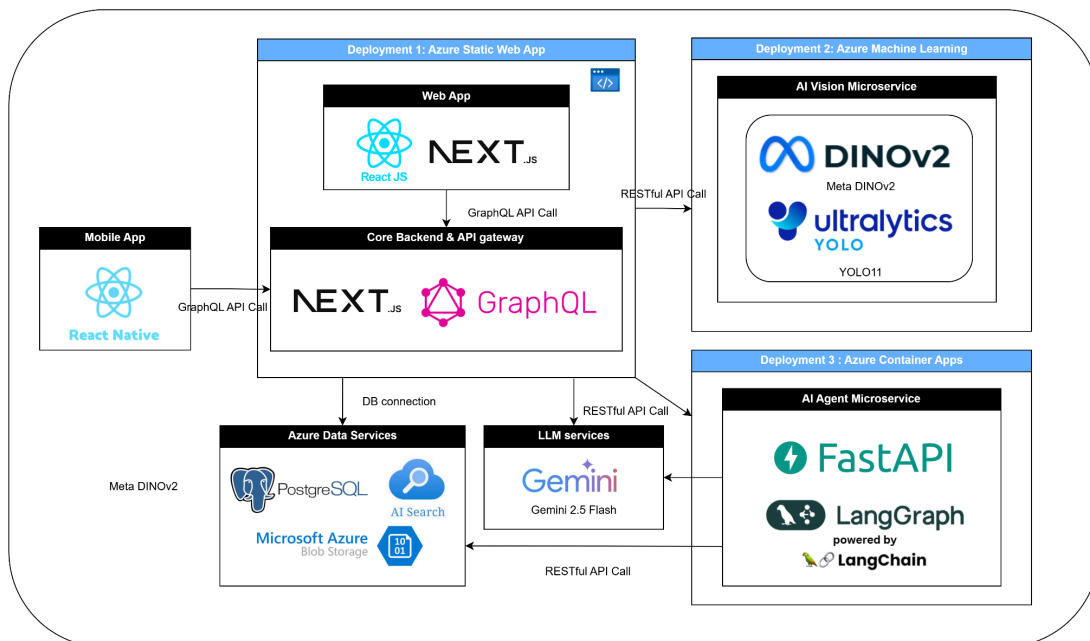


Figure 21 : PenangLens' Tech Stack Diagram

The architecture consists of the following key components:

Component	Description	Deployment	Technologies & Services
Clients	A React Native Mobile App for tourists and a Next.js Web App (admin dashboard).	Azure Static Web App	React Native, Next.js
Core Backend & API Gateway	Next.js app serving as Backend-for-Frontend (BFF) and API Gateway, handling authentication, business logic	Azure Static Web App	Next.js, GraphQL
AI Vision Microservice	Hosts computational vision models (DINOv2 for image recognition, YOLO11 for detail architecture detection)	Azure Machine Learning	DINOv2, YOLO11, RESTful API
AI Agent Microservice	A FastAPI service for itinerary planning, using LangGraph/LangChain for complex multi-step logic and connecting to data services.	Azure Container Apps	FastAPI, LangGraph, LangChain
Azure Data Services	Suite of managed services for data persistence and for vector similarity searches.	-	PostgreSQL, Azure Blob Storage, Azure AI Search
LLM Services	Integration with Google Gemini Flash Mode for complex reasoning and simpler tasks.	-	Google Gemini Flash Model, RESTful API

Table 27: Implementation details of PenangLens

3.7.1 Hardware Requirements

1. **Development Workstation:** A standard laptop with at least 16GB RAM to run mobile emulators and the local development environment.
2. **Cloud Server Infrastructure:** Hosted on Microsoft Azure, utilizing GPU-enabled compute instances for AI processing and standard web servers for the API gateway.
3. **Client Mobile Device:** Any modern smartphone (Android equipped with a camera, GPS, and an active internet connection)

3.7.2 Software & Tools

1. Programming Languages & Frameworks

- Mobile App: React Native (JavaScript/TypeScript)
- Web Dashboard & API Gateway: Next.js (React)
- AI Microservices: Python 3.10, FastAPI

2. AI Algorithms & Libraries

- Vision Model: DINOv2 (Feature Extraction), YOLO11 (Object Detection)
- Agent Framework: LangGraph & LangChain (Reasoning Loop)
- LLM API: Google Gemini Flash Mode (Generative Storytelling)

3. Development Tools

- IDE: Visual Studio Code
- Version Control: Git & GitHub
- Containerization: Docker
- API Testing: Postman
- Design: Figma

3.7.3 Data Source and Preparation Strategy

To ensure the system accurately interprets Penang's unique architectural context, a custom dataset is constructed to supplement pre-trained models. The data preparation follows two distinct pipelines:

1. Visual Dataset for Object Detection (YOLO11 Fine-Tuning)

To enable the detection of specific heritage elements (e.g., "Peranakan Tiles", "Kek Lok Si Pagoda architecture") that are absent in standard generic datasets (like COCO), the YOLO11 model undergoes domain-specific fine-tuning.

- **Data Acquisition:** A custom dataset will be curated by browsing online heritage archives and supplementing them with high-resolution photos manually captured at Georgetown sites.
- **Data Annotation:** The raw images undergo supervised annotation using **Roboflow**. Bounding boxes are manually labeled for specific architectural classes to create the "ground truth" data required for supervised training.
- **Data Augmentation:** To improve model robustness against real-world tourism conditions, the dataset is artificially expanded using augmentation techniques such as random rotation, color jittering (simulating different lighting), and horizontal flipping.
- **Dataset Splitting:** The final dataset is partitioned into Training (70%), Validation (20%), and Testing (10%) sets to rigorously evaluate model performance and prevent overfitting.

2. Textual Corpus for Retrieval-Augmented Generation (RAG)

To support the generative storytelling engine and mitigate hallucinations, the system utilizes a verified external knowledge base.

- **Content Curation:** Descriptive text, historical facts, and logistical data are manually aggregated from reputable sources and official tourism portals.
- **Vectorization:** This text is processed into chunks and embedded into the vector database. It serves not as training data, but as the "ground truth" context retrieved at runtime to validate the Gemini LLM's responses.

3.8 AI Evaluation Plan

To ensure system reliability, a rigorous evaluation strategy is employed for the two primary intelligent pipelines: Computer Vision (Visual Analysis) and Generative AI (Storytelling).

3.8.1 Computer Vision Evaluation Pipeline

The visual recognition system (DINOv2 and YOLO11) is evaluated using standard metrics against a stratified test set of Penang tourist attractions.

1. Data Splitting Strategy (The 70/20/10 Rule)

Due to data scarcity for niche landmarks, a Stratified Random Split strategy is used:

- **Training (70%):** Used for fine-tuning. Augmented with rotation, color jittering, and flipping to prevent overfitting.
- **Validation (20%):** Used for hyperparameter tuning during training.
- **Testing (10%):** A "held-out" set of original images used for final blind evaluation.

2. Key Performance Metrics

- **Metric 1: Landmark Identification (DINOv2)**
 - Measure: Top-1 Accuracy.
 - Target: > 85%. Measures how often the correct Landmark ID appears as the top 1 match in the Vector Search.
- **Metric 2: Detail Detection (YOLO11)**
 - Measure: Mean Average Precision (mAP@0.5).
 - Target: mAP > 0.75. Measures the precision and overlap of bounding boxes for architectural details (e.g., "Roofs", "Carvings" or even layers).

3.8.2 Generative AI & RAG Evaluation (Using Ragas)

The storytelling and planning modules rely on the Google Gemini LLM retrieving facts from the Azure Vector Database. The Ragas Framework is utilized to automatically score performance using a "Judge LLM."

1. Faithfulness (Hallucination Check)

- **Objective:** Ensure stories are factually grounded in the retrieved database context.
- **Method:** Measures if claims in the generated answer can be inferred directly from the retrieved context (Score 0-1).
- **Target:** > 0.9 (Critical for historical accuracy).

2. Answer Relevance

- **Objective:** Verify that the AI Agent's itinerary addresses specific user constraints (e.g., "History focus").
- **Method:** Evaluates the cosine similarity between the generated response and the original user prompt.
- **Target:** > 0.85.

3. Context Precision & Recall

- **Objective:** Evaluate the quality of the Vector Search (Azure AI Search).
- **Method:** Measures if the retrieved documents actually contain the "Ground Truth" answer required to respond to the user.

3.9 SDG Alignment

This project closely aligns with the United Nations Sustainable Development Goals (SDGs), primarily contributing to:

Goal 8: Decent Work and Economic Growth - Specifically sub-goal 8.9

By 2030, devise and implement policies to promote sustainable tourism that creates jobs and promotes local culture and products.

PenangLens directly contributes to this goal by enhancing the sustainability and economic value of Penang's tourism industry. The global travel and tourism sector is projected to contribute USD 15.5 trillion to the global economy by 2033, with technology-driven innovations playing a crucial role in this growth [34]. By providing an innovative digital tourism product, PenangLens can help Penang capture a larger share of the growing experiential tourism market, which is valued at USD 1.1-1.3 trillion globally [35]. The application promotes local culture and heritage through AI-powered storytelling, potentially increasing visitor engagement and length of stay, thereby generating more economic benefits for local communities and cultural heritage preservation efforts.

Goal 11: Sustainable Cities and Communities - Specifically sub-goal 11.4

Strengthen efforts to protect and safeguard the world's cultural and natural heritage.

This project directly contributes to protecting and promoting cultural heritage by innovating digital methods for cultural heritage dissemination and education.

UNESCO recognizes that cultural heritage sites face increasing pressures from tourism, urbanization, and environmental factors [36]. PenangLens addresses these challenges by providing sustainable digital alternatives to traditional heritage interpretation methods, reducing physical wear on heritage sites while simultaneously increasing cultural awareness and appreciation. The application's ability to distribute tourist attention across multiple heritage sites and architectural details can help alleviate overcrowding at popular landmarks, supporting more sustainable heritage tourism practices. Furthermore, the Personalized Itinerary Engine can be programmed to intelligently route tourists to lesser known but culturally significant sites, actively helping to distribute tourist flow away from overburdened hotspots and promoting a more balanced and sustainable exploration of the city's heritage.

4. CONCLUSION

In conclusion, this project addresses the challenges of information overload and superficial engagement in Penang's tourism sector by proposing PenangLens, an AI-powered mobile companion designed to deepen cultural understanding and streamline travel logistics. As modern travelers increasingly seek authentic, experiential journeys across Penang's diverse landscape, there is a critical need to bridge the gap between visual discovery and historical context using advanced technology. The proposed system leverages state-of-the-art computer vision and generative AI to introduce an integrated platform for instant landmark interpretation and proactive itinerary planning. By utilizing a novel "coarse-to-fine" recognition pipeline, users can seamlessly decode the specific details of their surroundings, while an autonomous AI agent resolves the "paradox of choice" by generating personalized, feasible travel plans in real-time. Furthermore, the inclusion of a "Human-in-the-Loop" administration system ensures content accuracy and scalability, positioning PenangLens as a sustainable, high-impact solution that significantly enhances the competitiveness of Penang's tourism industry.

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